

SECTION 7 – RF SHIELDED ROOM

TABLE OF CONTENTS

<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
7-1	RF SHIELDED ROOM SPECIFICATION	7-3
7-2	VENTS	7-4
	7-2-1 Cryogenic	7-4
	7-2-2 Determining Vent Location	7-4
	7-2-3 Waveguide	7-6
	7-2-4 Guide For Outside RF Room Isolation Joint	7-7
	7-2-5 HVAC	7-8
7-3	PLUMBING	7-9
	7-3-1 Water	7-9
	7-3-2 Medical Gases	7-9
	7-3-3 Sprinklers	7-9
7-4	ELECTRICAL	7-10
7-5	RF PENETRATION PANEL	7-14
7-6	PHYSICAL CONSIDERATIONS	7-14
	7-6-1 Doors and Other Openings	7-14
	7-6-2 Ceiling Height	7-16
	7-6-3 Walls	7-18
	7-6-4 Floors	7-18
7-7	MAGNET MOUNTING REQUIREMENTS INSIDE RF SHIELDED ROOM	7-20
	7-7-1 Overview & Background Information	7-20
	7-7-2 Roles & Responsibilities	7-21
	7-7-3 Slab On Grade Sites Physical Requirements	7-23
	7-7-4 Seismic Zone & Suspension Foundation Sites Physical Requirements	7-27
	7-7-5 RF Shield Integrity	7-34
	7-7-6 Electrical Isolation	7-34
	7-7-7 Magnet Mounting Plate Anchors Installation Location	7-35
7-8	MAGNET ROOM – WALL MOUNTED EQUIPMENT REQUIREMENTS	7-36
	7-8-1 Remote Oxygen Sensor Module (OM3) – Optional	7-36
	7-8-2 Magnet Rundown Unit (MS4)	7-36
	7-8-3 Emergency Off buttons	7-36

TABLE OF CONTENTS (Continued)

<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
7-9	RF DOOR SWITCH	7-36
7-10	EMERGENCY EXIT	7-36
7-11	ROOM VENTILATION SWITCH	7-36

7-1 RF SHIELDED ROOM SPECIFICATION

Every GE MR system requires that the Magnet Room be RF shielded. Table 7-1 contains the RF Shielded Room specifications.

Note

It is advisable to check the integrity of all RF Shielded (screen) Rooms in proximity to the MR system. Poor RF Shield integrity can contribute to system image quality artifacts i.e. zippers.

TABLE 7-1
RF SHIELDED ROOM REQUIREMENTS

PARAMETER	REQUIREMENTS
RF ATTENUATION	100dB (10MHz – 100MHz) planewave
GROUND ISOLATION	1,000 ohms or greater Refer to Section 7-7-6 Electrical Isolation for additional information on electrical isolation requirement.
MATERIALS See Note 1	The choice of material is the responsibility of the customer's architect and RF vendor. Normally, copper-brass or treated aluminum is used because these materials are non-magnetic and will not affect homogeneity. However, RF Shielding has also been fabricated from galvanized steel or by modifying steel magnetic shielding to produce the required RF attenuation. Any steel RF enclosure will affect the magnet's homogeneity and must be reviewed by GE Medical Systems MR Siting and Shielding Group. The door or any other moving or non-rigid parts must not be fabricated from magnetic materials.
CONSTRUCTION See Note 1	- The design of the shield support system is the responsibility of the customer's architect and RF vendor. If magnetic steel panels are used, these materials must be rigidly supported to prevent any slight movement, from air pressure changes or other reasons, that could degrade magnet homogeneity and system performance. For safety reasons, magnetic steel material must also be well anchored; loose steel components can become dangerous projectiles and accelerated into the magnet. - It is the customer's responsibility to coordinate magnet mounting methods with the RF shielded room vendor to prevent RF leaks and secondary grounding problems.
TESTING	- The customer's architect and RF vendor are responsible for conducting RF attenuation and ground isolation tests to verify that the shield meets GE specifications. The RF shielded room verification test is to be performed in the presence of a GE representative. - When to test: The RF shielded room must be 100% finished, magnet installed, all penetrations complete including floor mounting bolts. The test must be conducted with an RF vendor supplied blank penetration panel and the same mounting hardware which is used with the GE penetration panel. - How to test: The FINAL RF Shielded room acceptance test shall be performed in accordance with Appendix A, RF SHIELDED ENCLOSURE TEST GUIDELINE.
MAINTENANCE	Follow RF vendor's recommended maintenance. Alert GE Service Representative of any RF shielded room maintenance issues since there may be system performance impacts.
ACOUSTIC	RF Screen Room including all openings (i.e. windows, doors, vents, etc.) need acoustic properties to meet local regulations and customer requirements. Note RF Screen Room doors with < 45 to 50 db acoustic attenuation have caused customer acoustics issues. Refer to Section 4-7 ACOUSTICS for additional information.
Note 1 Suspension Foundation and Seismic Zone sites requires the Magnet Mounting Plate be installed into the Magnet Room floor and the plate must be utilized as the RF Shield under the Magnet. In cases where the RF Shield is located well below the magnet, i.e. in a combined RF and Magnetic shield design, than the magnet mounting plate will not be utilized as part of the RF Shield. Refer to Section 7-7 MAGNET MOUNTING REQUIREMENTS INSIDE RF SHIELDED ROOM.	

7-2 VENTS

7-2-1 Cryogenic

Due to normal boil-off of liquid helium and the possibility of a quench with superconducting magnets, outside cryogenic venting is required. RF shielded room contractor is to provide one straight pipe with maximum 0.125 in. (3.175 mm) wall thickness for the cryogenic vent pipe/waveguide. The vent pipe/waveguide is to be made of non-magnetic metal (e.g. aluminium or stainless steel) which is grounded to the RF room and electrically isolated from any other grounds. The vent pipe/waveguide must extend inside and outside of the RF shielded room to allow for non-metallic isolation joint connections. The HVAC (heating, ventilation, and air conditioning) contractor is to make cryogenic vent connections to vent pipe/waveguide outside of the RF shield and GE will make the normal connection in the Magnet Room. Refer to Section 4-9-2, Requirements For Inside Magnet Room, for exceptions.

7-2-2 Determining Cryogenic Vent Location

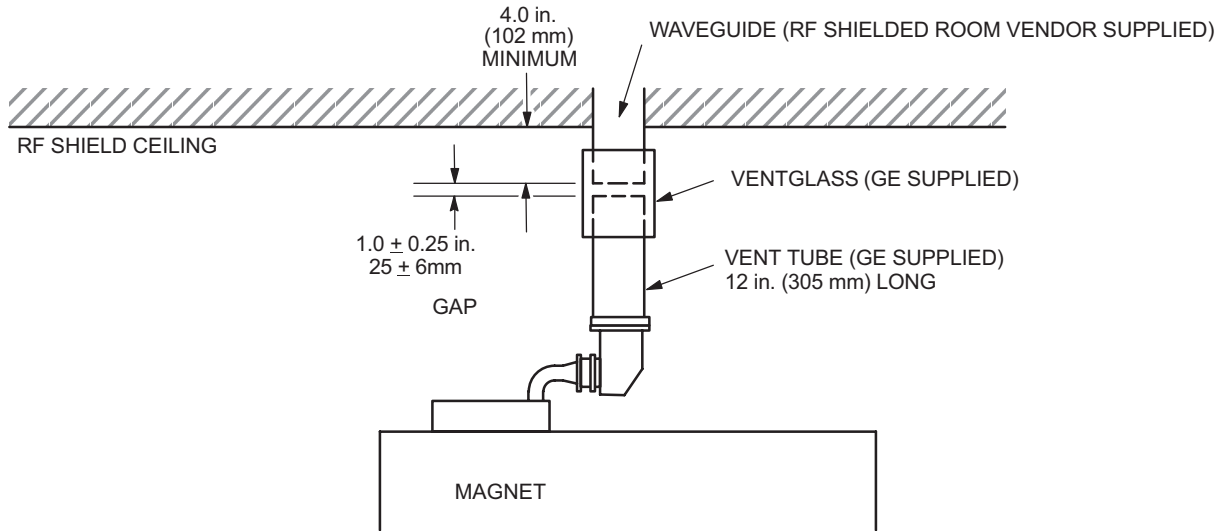
Shown in Illustration 7-17 is the 0.7T magnet vent location relative to isocenter on the magnet mounting plate. It is important that the 0.25 in. (6.35 mm) tolerance for vent opening be met. Included with the basic Signa system is an 8 in. (203.2 mm) ID vent tube. A typical route for this vent tube is shown in Illustration 7-1. However, the customer's contractor is responsible for the design and installation of a cryogenic vent system which meets the requirements in Section 4-9, CRYOGENIC VENTING.

GE provides a Floating Flange Vent Adaptor Kit with the 0.7T magnet. The kit increases the vent adapter radial adjustability in any direction to one inch (25.4 mm). The adjustability helps absorb some mismatch (1 inch (25.4 mm) in any horizontal direction) of the locations of the Vent Pipe from the magnet and the Ceiling Vent Pipe, installed by the RF Room contractor. See Illustration 7-2. The GE provided Vent Pipe is 12 inches (305 mm) long with a wall thickness of 0.125 inch (3.175 mm) and may be cut short to create 1 ± 0.25 inch (25 ± 6 mm) gap between the Vent Pipe and the Ceiling Waveguide for dielectric isolation to ensure the integrity of the RF shield room.

7-2-2 Determining Cryogenic Vent Location (Continued)

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.

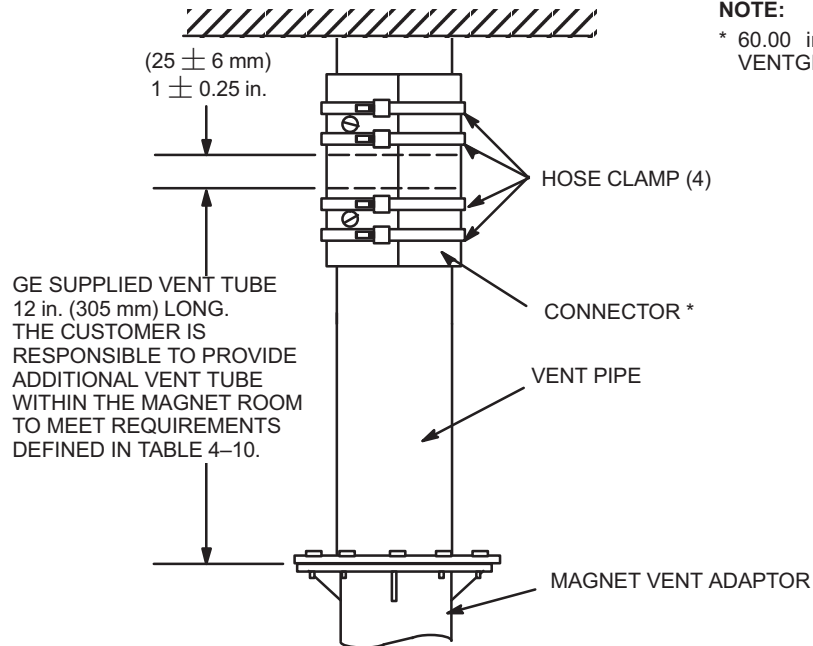


CRYOGENIC VENT ROUTING

ILLUSTRATION 7-1

NOTE:

- * 60.00 in X 8.00 in (152 cm X 20 cm)
VENTGLASS CONTINUOUS WRAP SLEEVE



FLOATING FLANGE

ILLUSTRATION 7-2

7-2-3 Waveguide

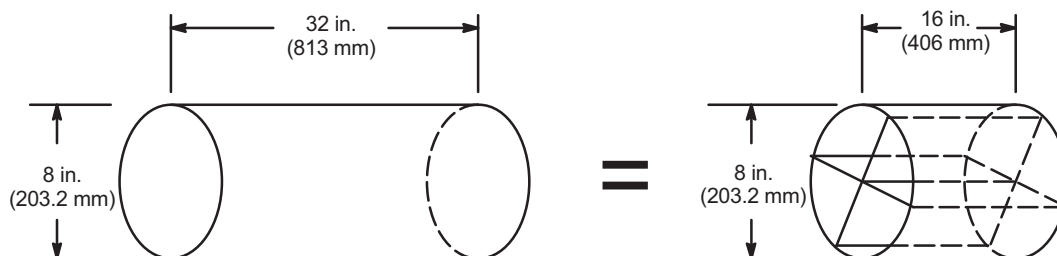
RF shield room contractor/designer is responsible for choosing and installing a RF shield waveguide . The generally accepted length of the waveguide is four times the outside diameter of the tube. Therefore, a 8 in. (203.2 mm) OD waveguide should be 32 in. (813 mm) long. Refer to Table 7-2 for list of GE requirements for the waveguide.

TABLE 7-2
WAVEGUIDE REQUIREMENTS

PARAMETER	REQUIREMENTS
WAVEGUIDE SIZE	- If the provided GE vent tube is to be connected directly to the waveguide via the GE Ventglass method, the outside diameter of the waveguide must match the outside diameter of the GE vent tube within ± 0.125 in. (3 mm). Larger mismatches in diameters will result in unacceptable leakage during a quench. - The generally accepted length of the waveguide is four times the inside diameter of the tube. Therefore, an 8 in. (203.2 mm) OD waveguide should be 32 in. (813 mm) long.
WAVEGUIDE MATERIAL	The Waveguide must be constructed from one of the GE accepted materials (i.e. stainless steel, aluminium or copper). Typically, the waveguide is made from the same material as the RF shielded enclosure to avoid dissimilar metal interfaces (i.e. galvanic reaction).
WAVEGUIDE CONSTRUCTION	- The waveguide does not have to be positioned equally on either side of the RF ceiling (or wall). For example, 8 in. (203.2 mm) of a 32 in. (813 mm) long waveguide may hang down below the RF ceiling with the remaining 26 in. (660 mm) extending above. - If a 90 degree elbow is required to avoid ceiling structures, the elbow can be a part of the waveguide and contribute to its overall length. Waveguides do not have to be completely straight. - If a full length waveguide is not possible due to structural interferences, a shorter waveguide can be fabricated by dividing the inside volume into no more than four chambers. For example, if an 8 in. (203.2 mm) OD waveguide is divided into four equal chambers, as shown in ILLUSTRATION 7-3, the length of the waveguide may be decreased from 32 in. (813 mm) to 16 in. (406 mm). If this technique is used, 1 psig must be added to the pressure drop calculation to account for the pressure drop of the four chambered waveguide. (Section 4-9, CRYOGENIC VENTING) - Flat, honeycomb type waveguide is not acceptable.

NOTE:

- 1 psi MUST BE ADDED TO THE PRESSURE DROP CALCULATION TO ACCOUNT FOR THE PRESSURE DROP OF THE FOUR CHAMBERED WAVEGUIDE.
- IN A CASE OF WAVEGUIDE LENGTH RESTRICTION, A HALF LENGTH WAVEGUIDE WITH FOUR CHAMBERS MAY BE USED.



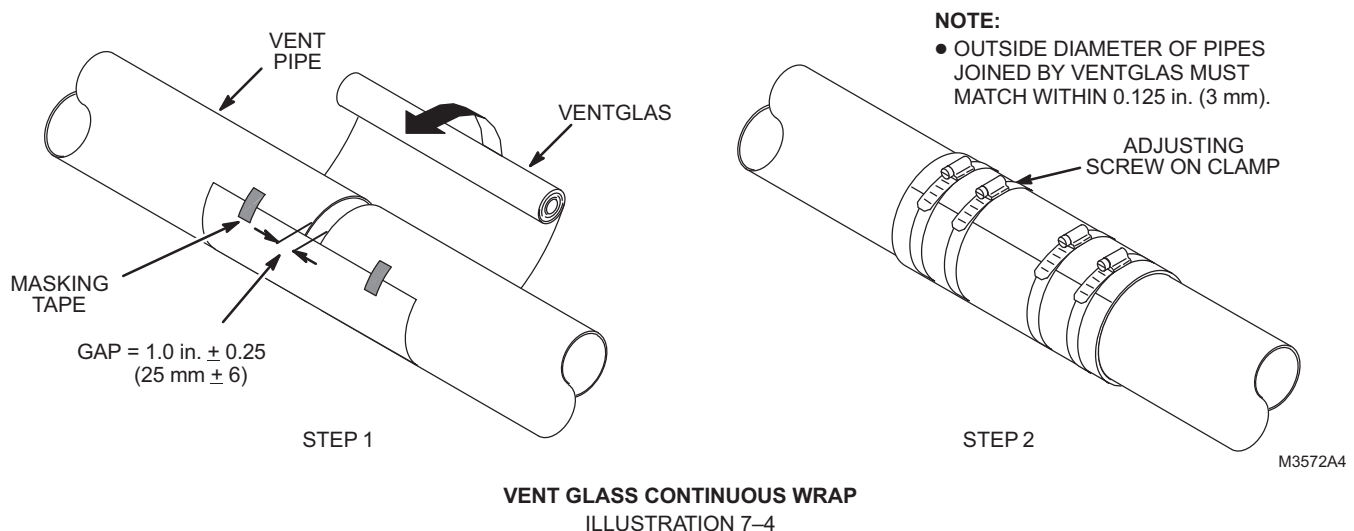
4 CHAMBER WAVEGUIDE
ILLUSTRATION 7-3

7-2-4 Guide For Outside RF Room Isolation Joint

The RF shielded room contractor/designer is responsible for choosing and installing an isolation joint outside of the RF shielded room as shown in Section 4-9-1, Requirements For Outside Magnet Room. This isolation joint is required to maintain the single point ground concept for the RF shielded room. Table 7-3 contains suggestions for the RF room isolation joint.

TABLE 7-3
OUTSIDE RF ROOM ISOLATION JOINT SUGGESTIONS

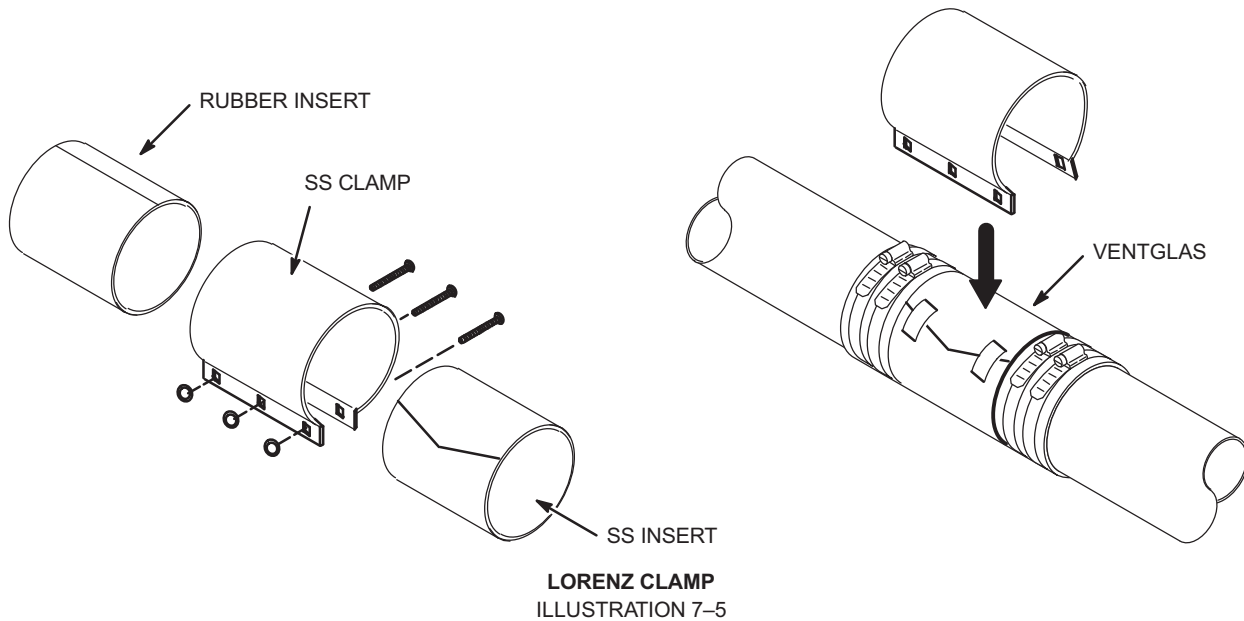
PARAMETER	ISOLATION JOINT SUGGESTIONS
ISOLATION JOINT MATERIAL	- PVC, rubber or soil pipes must not be used to construct the isolation joint. - Ventglas and Lorenz clamp are two GE recommended methods of achieving the isolation.
ISOLATION JOINT CONSTRUCTION	- Ventglas: If the connection diameter is 8 in. (203.2 mm), a Ventglas connection method as shown in Illustration 7-4 is recommended. - Lorenz clamp: If the connection diameter is 8 in. (203.2 mm), a Lorenz clamp connection as shown in Illustration 7-5 is recommended. - The mating diameters must match within ± 0.125 in. (3 mm). - The Ventglas must not be used for structural support.
INFORMATION	- Ventglas information may be obtained from: Industrial Machine & Fabricating Inc. 2808 E Sammys Lane Florence, SC 29506-3841 USA (843) 667-4582 e-mail address indmachfab@aol.com web site address www.ventfabrics.com - Vent Fabrics Inc. 5520 N Lynch Avenue Chicago, IL 60630-1418 USA (800) 621-1207 or (773) 775-4477 - Lorenz clamp information may be obtained from: Lorenz and Son Mfg. Co. LTD. P.O. Box 1002 Cobourg, Ontario, Canada K9A4W4 (905) 372-2240, fax (905) 372-4456



7-2-4 Guide For Outside RF Room Isolation Joint (Continued)

NOTE:

- LORENZ CLAMP IS ATTACHED OVER THE VENT GLASS FOR ADDED REINFORCEMENT.
- SS = STAINLESS STEEL



7-2-5 HVAC

RF shielded room contractor is to install HVAC waveguides (open pipes or honeycomb-type) which penetrate room and to ensure waveguides are non-magnetic and electrically isolated. HVAC contractor is to determine size and number of vents, consistent with local codes.

An exhaust fan placed outside the RF shielding with appropriate wave guide filtering is required for quick removal of helium gas in the event large amounts of helium disperse into the Magnet Room. The exhaust fan can be connected to the output relay of the optional oxygen monitor. The fan will then be activated in the event the room oxygen level is less than 18%. Refer to Section 4-8, ROOM VENTILATION, for other exhaust fan requirements.

7-3 PLUMBING

All metallic pipes entering the RF Room, excluding cryogenic vent and floor drains, must be located within 30 inches (762 mm) of the RF common ground.

Note

When welding in an MR room with system equipment installed, the return path for the welding must be in very close proximity to the welding. The close proximity is needed to make sure the welding currents do not cause damage to the system. Never use the building structure as a return path for welding.

7-3-1 Water

All pipe waveguides must meet the 100 dB requirements. If a floor drain is to be installed in the Magnet Room, it must be electrically isolated and meet the 100 dB requirements. All plumbing must be consistent with local codes.

7-3-2 Medical Gases

The customer should consider if medical gases are to be piped into the Magnet Room along with suction service for patient life support. Remember, all non-electrical entries into the Magnet Room must use appropriate waveguide. Special precaution must be taken to ensure that ferromagnetic medical gas cylinders are not brought into the Magnet Room.

7-3-3 Sprinklers

If using sprinklers in the Magnet Room, dry pipe systems have the advantage of reducing ground problems. However, all decisions regarding fire protection systems are the customer's responsibility. If wet-type sprinkler system is used, pipe penetration should be limited to one location.

7-4 ELECTRICAL

The entry of any electrical lines into the RF shielded room must be filtered to ensure that the RF shielded room meets the minimum attenuation levels. The RF shielded room vendor must supply filters for all penetrations of the RF shielding excluding the lines entering through the GE supplied RF penetration panel. All filters (for electrical lines) must be located outside the 200 gauss line.

Note

AC outlets must have ground wires firmly attached per electrical codes to avoid intermittent grounding which can cause undesirable emi (electro magnetic interference) issues within the RF shielded room.

RF shielded room vendor should review with the electrical contractor the number of incoming power lines to the Magnet Room in determining the number of filters needed for electrical requirements.

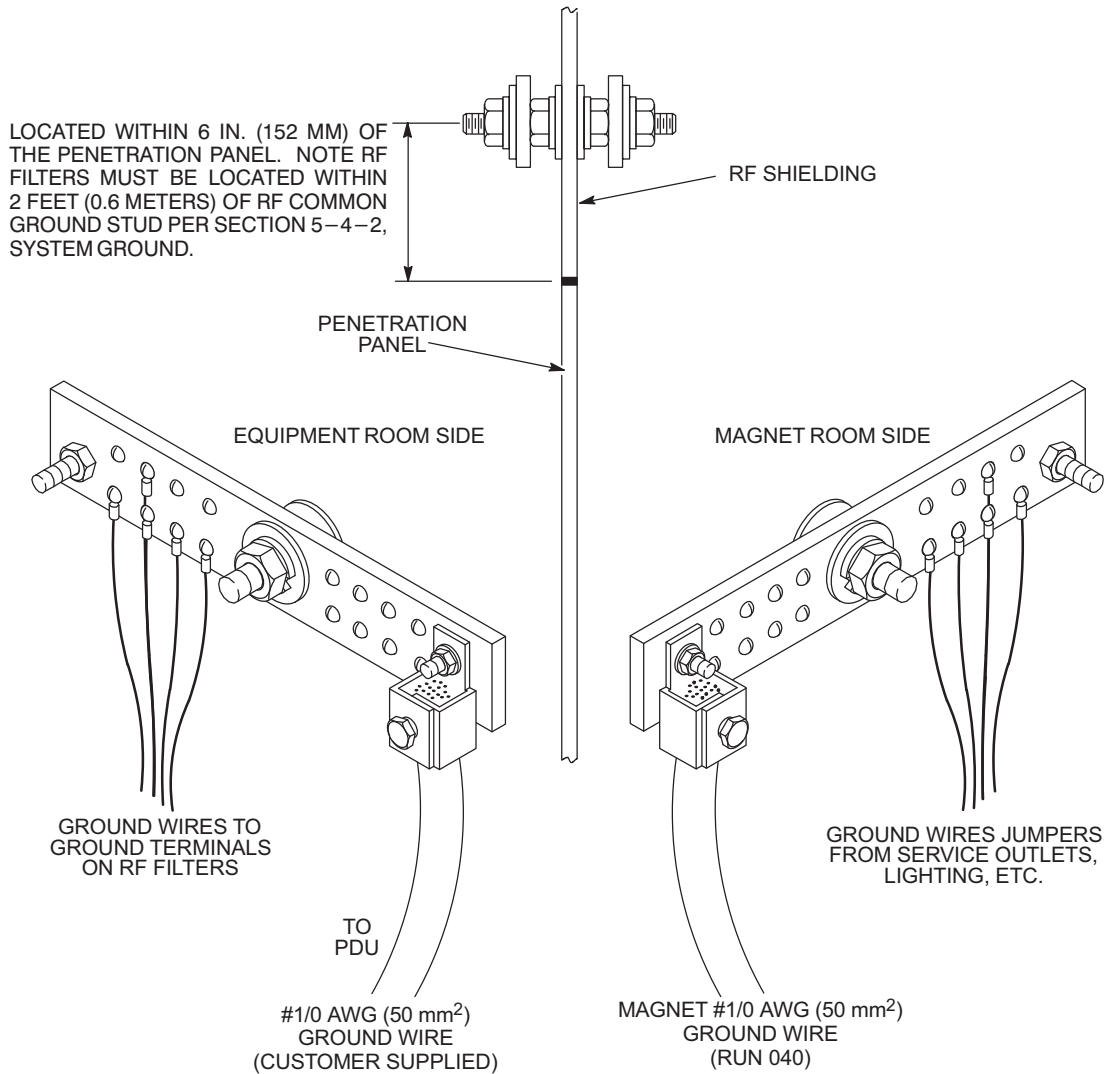
The power ground lines from customer supplied power filters must be grounded to the RF shield common ground point and be located as close as possible to the RF penetration panel. See Section 5, POWER REQUIREMENTS, for power and grounding requirements of all incoming power lines to the RF shielded room.

All lighting in the RF shielded room must be DC/incandescent. Dimmer switches must not be used; however, a selectable switch may be used to change the light intensity. For additional Magnet Room lighting information refer to Section 4, SITE ENVIRONMENT, Section 5, POWER REQUIREMENTS, and Section 6, INTERCONNECT DATA.

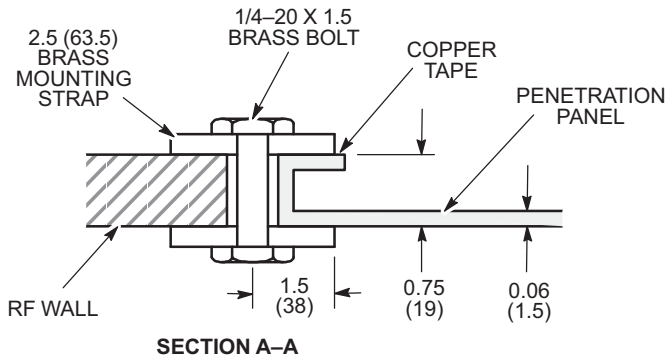
Common ground connection for shielded room must be located within 6 in. (152 mm) of the RF shielded room Penetration Panel with RF filters located within 2 feet (0.6 meters) of the RF common ground. RF shielded room vendor to provide this common ground connection on both sides of shielded room by means of a stud extending through the shielded room (see Illustration 7-6). RF Common ground stud must be accessible for servicing purposes on both sides of shield room. For aesthetic purposes, it is recommended that the stud be positioned above the Penetration Panel so it is concealed behind the penetration panel cover (see Illustration 7-9).

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS

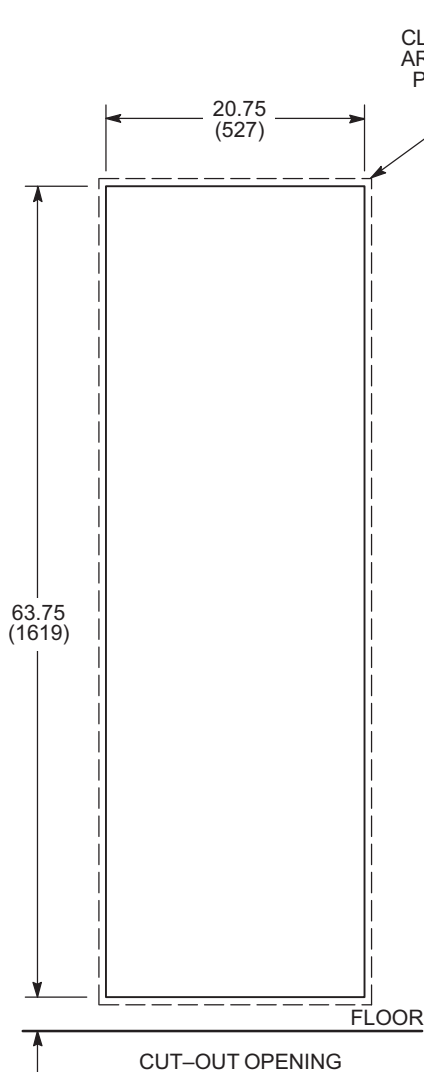


RF COMMON GROUND PENETRATION STUD
ILLUSTRATION 7-6

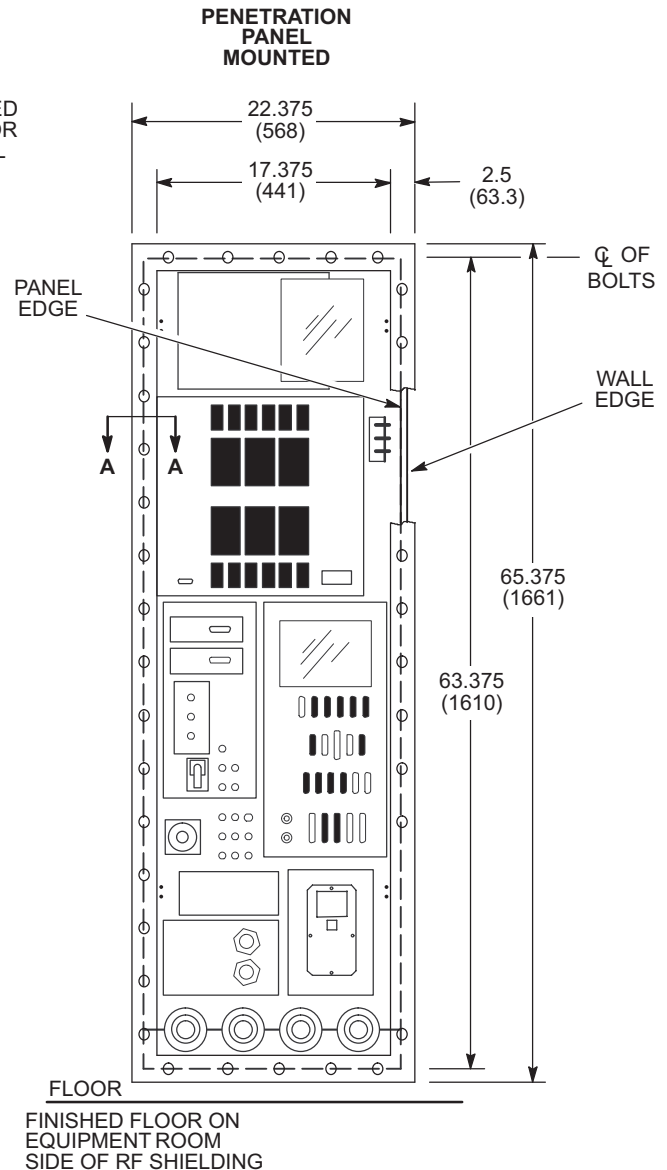


NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- ALL DIMENSIONS ARE ± 0.0625 (1.6)

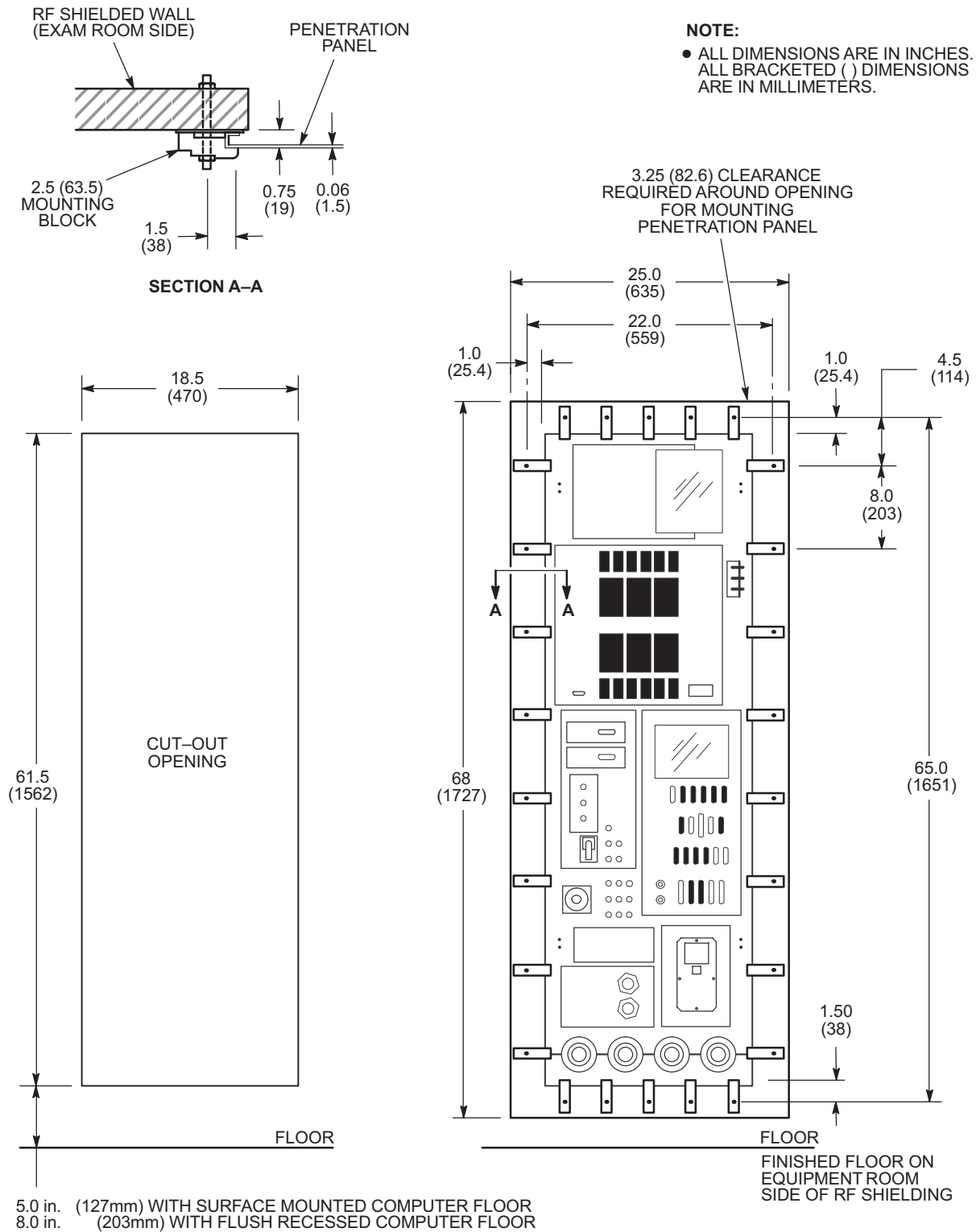


3.0 in. (76 mm) WITH SURFACE MOUNTED COMPUTER FLOOR
6.0 in. (152 mm) WITH FLUSH RECESSED COMPUTER FLOOR



PENETRATION PANEL CUT OUT FOR 0.75 INCH (19 MM) THICK RF WALL
ILLUSTRATION 7-7

M4526A



M4527A

PENETRATION PANEL CUT OUT FOR RF WALL THICKNESS VARYING FROM SMALL TO LARGE
ILLUSTRATION 7-8

7-5 RF PENETRATION PANEL

Illustrations 7-7 and 7-8 show two different methods for mounting the GE MR penetration panel. Either method may be used depending on RF shielded room wall thickness. Ensure if the mounting method in Illustration 7-7 is used, the RF wall thickness is 0.75 in. (19 mm) $\pm$ 0.0625 in. (1.6 mm). Check with RF shielded room vendor to determine appropriate mounting method. The penetration panel must be covered on both sides for safety. If GE supplied adjustable covers are not used, customer must furnish covers or enclosures with key or tool required for opening to limit access to the panel.

The penetration panel is to be mounted above the finished floor on the penetration cabinet side of the RF shielded room. GE supplies only the penetration panel as shown in Illustrations 7-7 and 7-8. The mounting and clearance dimensions for the penetration panel covering are shown in Illustration 7-9.

RF shielded room vendor must provide the opening in the RF shielding and appropriate mounting hardware for the GE penetration panel. The RF shielded room acceptance test must be performed after the opening is cut in the RF shielding for the GE penetration panel. This acceptance test must be conducted with vendor supplied blank panel and the same mounting hardware to be used with the GE penetration panel. It is the facility's responsibility to ensure that the RF shielded room vendor testing meets the attenuation specifications listed in Section 7-1, RF SHIELDED ROOM INTRODUCTION.

7-6 PHYSICAL CONSIDERATIONS

The RF shielded room can be either a free standing shielded structure or a shielded room within an existing room. Either style must be electrically isolated from earth ground by 1000 ohms or greater. Magnetic shields that are used as RF shield must also be isolated from ground by the same specification.

7-6-1 Doors and Other Openings

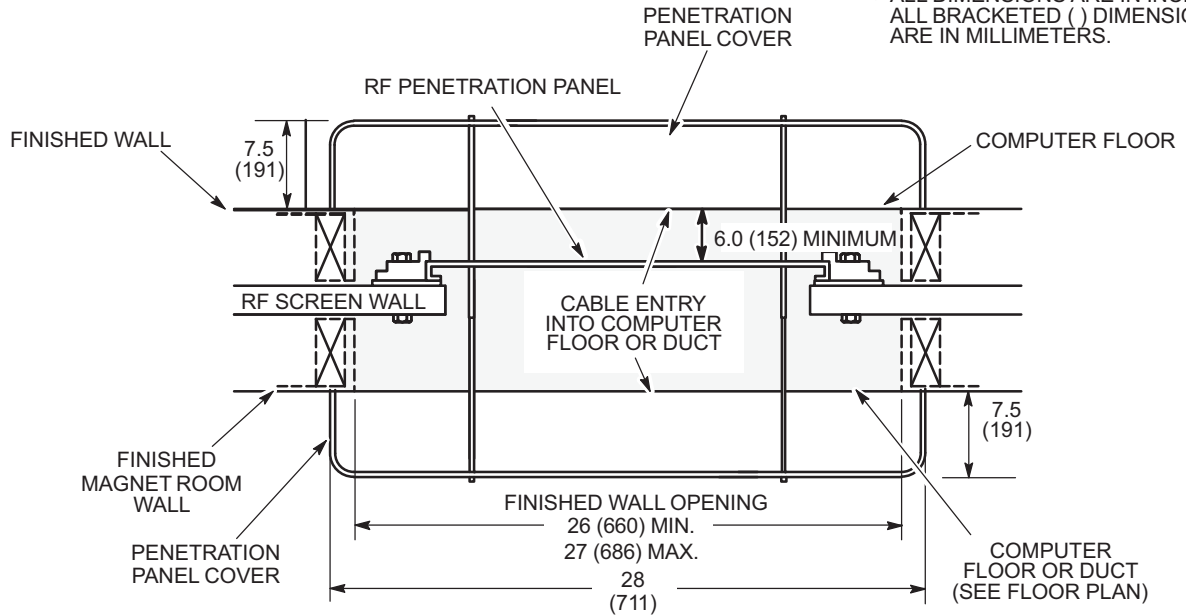
Shielded room doors are a major source of RFI leaks and should be kept to a minimum.

The main door requires a minimum finished opening of 43 in. (1092 mm) to allow for helium dewars and patient transports to pass through the opening. However, a 48 in. (1219 mm) wide door is recommended for easy maneuvering of the patient transports i.e. gurneys. Maximum door sill height is 1 in. (25 mm) with a 10 degree maximum threshold inclination.

■ For moving the magnet into the room, a 9 ft wide by 9 ft high removable panel wall or 9 ft by 9 ft roof hatch is required. This normally consists of several panels that are bolted together to ensure a good RF shield when opening is not in use. Since future access may be required for magnet entrance/exit through the removable panels, avoid permanently sealing the removable panels. Note the room removable wall panel or roof hatch opening may need to be larger to achieve the required opening defined above in this section. Consideration for clear opening dimensions is especially applicable to sites requiring magnetic shielding.

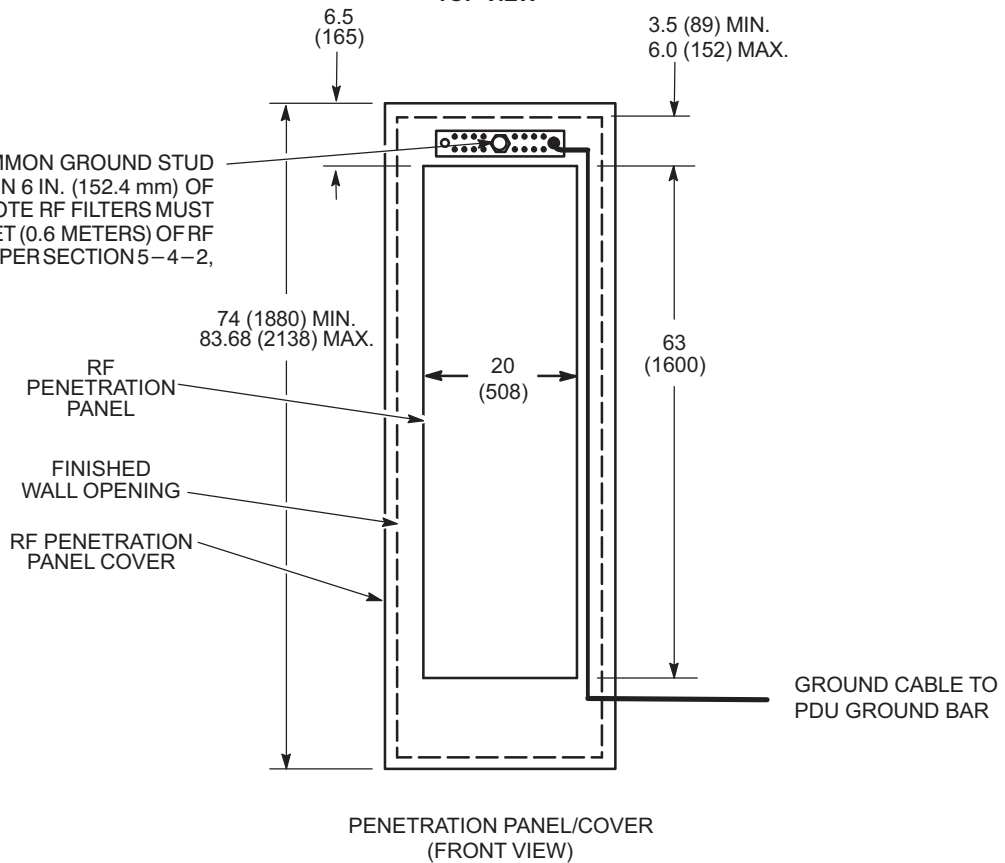
NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



TOP VIEW

RF SHIELDED ROOM COMMON GROUND STUD MUST BE LOCATED WITHIN 6 IN. (152.4 mm) OF PENETRATION PANEL. NOTE RF FILTERS MUST BE LOCATED WITHIN 2 FEET (0.6 METERS) OF RF COMMON GROUND STUD PER SECTION 5-4-2, SYSTEM GROUND.



PENETRATION PANEL/COVER (FRONT VIEW)

PENETRATION PANEL/COVERING MOUNTING REQUIREMENTS

ILLUSTRATION 7-9

7-6-2 Ceiling Height

Table 7-4 lists the absolute minimum (lowest) ceiling height required for servicing the 0.7T magnet, this height is required for the shaded area shown in Illustration 7-10. The cryogen vent must not be positioned or routed between the Magnet/Enclosure and the minimum (lowest) ceiling height in the areas indicated in Illustration 7-10 to allow for service access. GE Medical Systems, Modality Installation Planning Service group must be notified of any ceiling dimensions less than ceiling heights stated in Table 7-4 in the shaded area in Illustration 7-10. Modality Installation Planning Service group can be reached at (262) 548-4500.

TABLE 7-4
MAGNET SERVICING CEILING HEIGHT REQUIREMENTS

MAGNET TYPE	ABSOLUTE MINIMUM CEILING HEIGHT IN. (MM) See Note 1	COMMENTS
0.7T Magnet	102.68 (2608)	Magnet servicing is performed from a platform ladder which is positioned at the Coldhead side of the Magnet. Ceiling height allows clearance for ramp lead insertion, cryogen fill line stinger insertion, and coldhead extraction tool.
Note 1 Absolute minimum ceiling height values are from magnet room finished floor to fixed ceiling.		

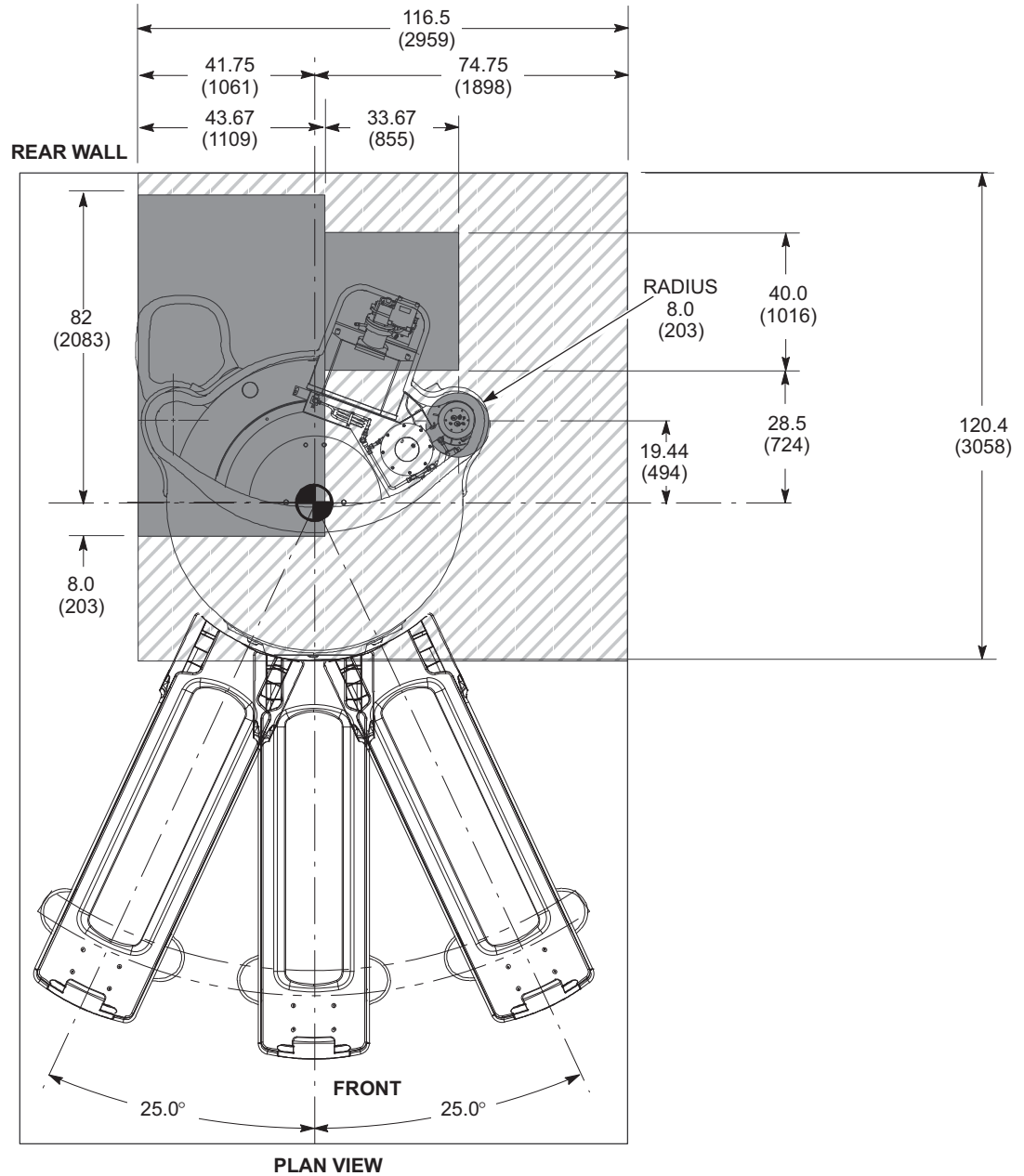
Use of a standard valved helium transfill line and a 250 liter dewar (not more than 70 in. (1778 mm) high) requires a ceiling height of 135.5 in. (3442 mm) for inserting transfill line into the dewar. Note that this need only be a 24 in. (610 mm) square ceiling recess located either in the Magnet Room or in an accessible area near the Magnet Room where the transfill line can be inserted into the dewar. A 500 liter dewar (not more than 73 in. (1854 mm) high) requires a ceiling height of 138 in. (3505 mm) for the same process.

If the helium transfill requirements cannot be satisfied in or near the Magnet Room, consider a location outside the building or on a loading dock. The standard valved transfill line, after insertion into either a 500 or 250 liter dewar, will fit through 79 in. (2007 mm) high doorways and hallways. Provide free access from the dewar location to the magnet. If elevators are to be used along cryogen delivery route, verify that elevator dimensions and weight capacity is sufficient to handle the cryogen dewars. Also, elevator must be dedicated with restricted access during cryogen transport (will not allow stops between initial start and final floor destination).

7-6-2 Ceiling Height (Continued)

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
-  SHADED AREA ROOM CEILING MUST NOT BE BELOW TABLE 7-4 REQUIREMENTS.
-  SHADED AREA INDICATES AREAS WHERE NOTHING CAN BE POSITIONED OR ROUTED BETWEEN MAGNET/ENCLOSURE AND MINIMUM CEILING HEIGHT. CRYOGEN VENT MUST NOT BE ROUTED IN THESE AREAS TO ALLOW FOR SERVICING ACCESS.



MINIMUM CEILING HEIGHT 0.7T MAGNET
ILLUSTRATION 7-10

7-6-3 Walls

It is recommended that walls be covered to protect RF material and to add to the aesthetics of the room for patient comfort. Fire retarding material must be used per building codes. Consult RF shield room vendor for RF shielding service requirements prior to covering RF walls. Removable wall covering may be needed if periodic RF shield servicing is required to maintain RF integrity.

The recommended patient viewing window dimensions are 48 in. wide by 42 in. high (1219 mm x 1067 mm). The location of the window is dependent on the position of Operator Workspace position.

Note

The operator at the Operator Workspace must be able to view the patient during a scan.

7-6-4 Floors

For Slab On Grade sites the concrete slab on which the magnet will be mounted must be continually supported on grade (no local voids allowed) and must be isolated from the rest of the building. Minimum concrete slab thickness must be 18 inches (457 mm) within the entire floor area of the RF Shield Room and cannot be limited to the magnet footprint area only. Refer to **Section 7-7 MAGNET MOUNTING REQUIREMENTS INSIDE RF SHIELDED ROOM** for specifications for mounting the magnet.

For Suspension Foundation sites the magnet mounting plate must be installed recessed into the floor. The magnet mounting plate must be utilized as the RF Shield under the Magnet and the magnet mounting plate must be mounted to the 18 inch (457 mm) concrete floor slab. In cases where the RF Shield is located well below the magnet, i.e. in a combined RF and Magnetic shield design, then the magnet mounting plate will not be utilized as part of the RF Shield. Refer to **Section 7-7 MAGNET MOUNTING REQUIREMENTS INSIDE RF SHIELDED ROOM** for specifications for mounting the magnet and the magnet mounting plate.

The Magnet Room floor levelness requirement is important for operation of the Patient Swing Table. Floor levelness in the Magnet Room must not exceed 0.236 in. (6 mm) between depressions and high spots. The floor itself must be no higher than the surface which the rear Patient Swing Table wheels contact, see shaded area shown in Illustration 7-11.

Note, the critical dimension of 41.34 in. ± 0.118 in. (1050 mm ± 3 mm) from finished floor to vertical center of the Magnet and Patient Swing Table rear wheel plate must be maintained. This will allow the Patient Swing Table to swing without obstruction from the floor.

Note

The Magnet foot/mounting pad must be in direct contact with the Magnet Mounting Plate. Do not use finished floor material under the Magnet foot flange or Patient Swing Table wheel plate to achieve final Patient Swing Table rear wheel plate alignment to Magnet vertical isocenter, refer to Illustration 7-18.

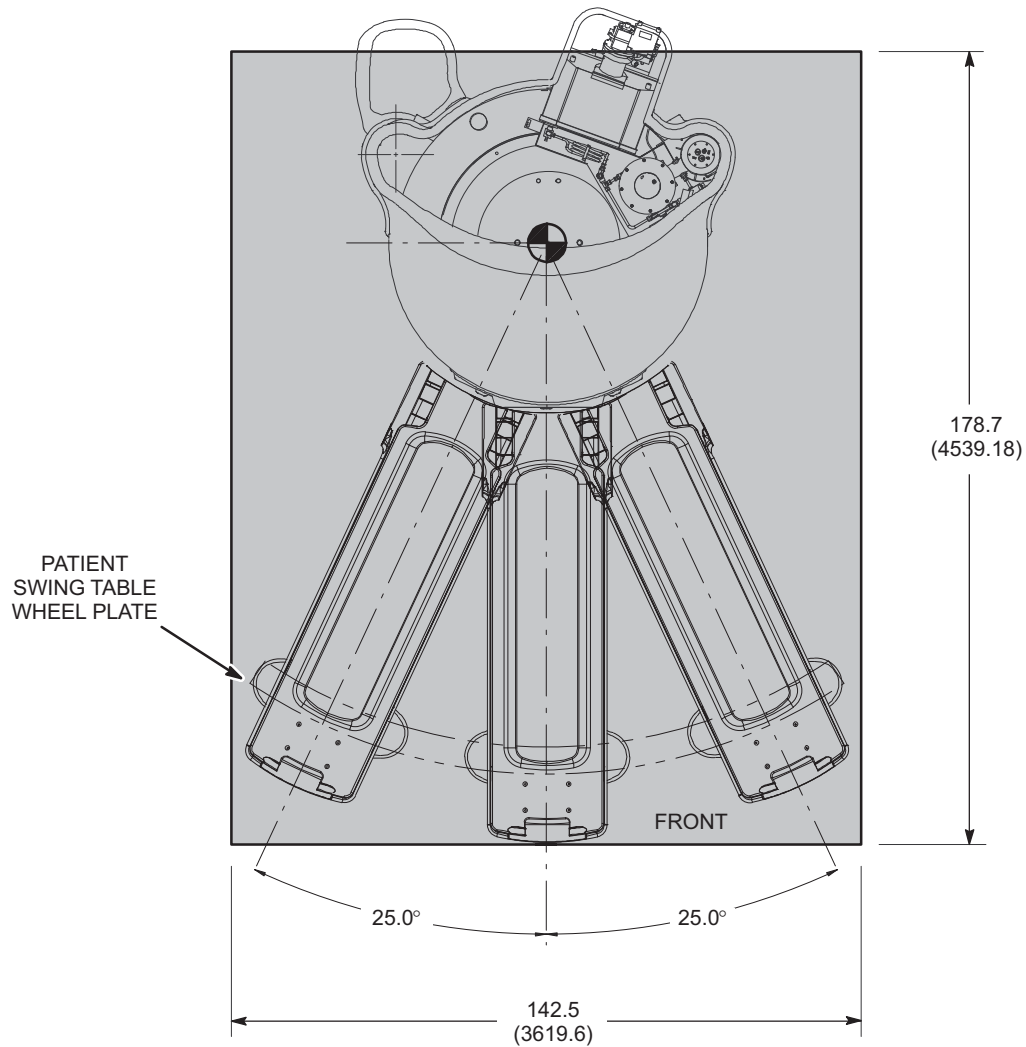
Magnet Room layout and RF Shield designs must include cable routing to the Magnet. Refer to Illustration 7-19 for cable access opening through the magnet mounting plate if customer plans to utilize cable access below the magnet.

The finished floor in the Magnet Room should be waterproof and be a conductive type flooring to reduce the possibility of a static discharge. The floor covering must be hard-surfaced with the seams sealed to protect the RF flooring against possible water damage. This floor should also be easily replaceable to repair tiles that are damaged from occasional cryogen spillage. Removable access flooring in the Magnet Room must be aluminium or non magnetic for safety purposes.

7-6-4 Floors (Continued)

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.



PLAN VIEW

MAGNET FLOOR LEVELNESS AREA

ILLUSTRATION 7-11

7-7 MAGNET MOUNTING REQUIREMENTS INSIDE RF SHIELDED ROOM

7-7-1 Overview & Background Information

All GE MR systems require an RF enclosure which is commonly called an RF Shielded Room or an RF Screen Room. RF signals from sources outside of the RF Shielded Room are attenuated so they do not interfere with the MR system. Likewise, the RF signals produced by the MR scanner are kept from interfering with other RF devices. This RF quiet environment is necessary for the MR system to produce quality images.

RF Shield Rooms come in a variety of shapes and sizes but the feature they all have in common is a total RF shield produced by one continuous ground plane. This is achieved by making the magnet mounting plate, walls, floor, doors and ceiling out of an electrically conductive material such as copper, aluminum, brass, or steel. All the room components (the magnet mounting plate, walls, floor, ceiling, etc.) must be electrically bonded together to form one solid, common ground plane. Once this is established the ground plane is then tied to earth ground to create the RF shield. This ground point is known as the PRIMARY ground. The primary grounding technique works well for any battery powered devices that may be operated within the RF Shield Room. Devices requiring facility power or the introduction of facility power into the Screen Room requires a change in the primary grounding technique for the RF Shield Room. The RF Shield Room must now be grounded back to the facility power ground.

The addition of water systems or other grounds required by the national electrical code will cause the RF Shield Room ground impedance to be ZERO ohms. These additional grounds that connect the outside of the RF Shield Room to earth grounds are called SECONDARY grounds. It is the secondary grounding that needs to be controlled. If the secondary ground introduces any current to the RF Shield, indicating the RF Shield is a better ground path than the secondary ground, then the current can set up electrical fields on the surface of the RF Shield which may cause image artifact.

The equipment anchors are generally installed into the grade below the RF Shield Room because the floor thickness within the room does not provide the proper embedment depth. With this in mind the hardware used to anchor the equipment will be either a bolt or a stud. This hardware must penetrate the RF Shield and be in full contact with the shield thereby preventing RF leaks at the point of penetration as specified in **Section 7-7-5 RF Shield Integrity**. In this case, the anchoring hardware may be in direct electrical contact to the anchored equipment. The anchoring device that is set below the RF Shield Room floor must be isolated from ground. If it should come in contact with rebar or wire mesh and this in turn is contacting building steel then a secondary ground has been established and needs to be corrected. The magnet mounting plate is supplied with alternate mounting holes to assist in the avoidance of hidden rebar.

RF Shield Rooms that have the RF shield far enough below the finished floor in the room to allow the anchor to be installed without penetrating the shield have no issues with secondary grounding.

7-7-2 Roles & Responsibilities

The roles and responsibilities of the Customer RF Shield Room vendor, and GE Medical Systems are detailed below.

The Customer is responsible for the following tasks:

- Coordinate equipment anchor location with the contracted RF shield room vendor, structural engineer, and architect to prevent RF leaks and secondary grounding problems.
- Coordinate with the contracted RF shield room vendor, structural engineer, and architect the design of the Magnet Room floor per GE requirements in preparation for Magnet Floor Preparation Kit or Magnet Mounting Plate installation, refer to information of this section and Section 7-6-4 Floors.
- Obtain any and all approvals necessary for the construction of equipment support and seismic anchoring. Provide copy of building inspector's (inspection) report and approval on anchor method to GE Medical Systems local office.
- Customer's mechanical contractor/rigger is responsible for the preparation of room floor and installation of the Magnet Floor Preparation Kit or the Magnet Mounting Plate.
- Customer's mechanical contractor is responsible for installation of the Magnet Floor Preparation Kit or the Magnet Mounting Plate anchors and the coldhead mounting anchors to meet specified torque.
- Customer's mechanical contractor to make sure the final finished height of anchor stud to be flush or below top surface of Magnet Floor Preparation Kit or the Magnet Mounting Plate.

The RF Shield Room vendor is responsible for the following tasks:

- Integration of the Magnet Floor Preparation Kit or the Magnet Mounting Plate into the RF Shield Room floor design.
- For Slab On Grade Sites: Layout and installation of Magnet Floor Preparation Kit and table rail anchors utilizing template supplied in kit. Coordination with Building Contractor/Architect for proper floor preparation to prevent interference with rebar or structural steel that would cause a secondary ground path through the anchor.
- For Suspension Foundation Sites: Layout and installation of the magnet mounting plate anchors (mounting plate can be utilized as template). Coordination with Building Contractor/Architect for proper floor preparation to prevent interference with rebar or structural steel that would cause a secondary ground path through the anchor.
- Participate in the installation of the magnet mounting plate to insure proper integration with RF Shield floor.
- Perform ground impedance test on installed anchors prior to connection to RF Shield floor.
- Perform RF integrity test on room after anchors and studs are installed and connected to RF Shield.
- Provide copy of ground impedance and RF room integrity tests to GE Medical Systems local office.
- All magnet mounting plate holes have been cleared of any debris and covered with RF tape prior to magnet positioning on mounting plate. Metal filings left in the holes may introduce image artifacts.

7-7-2 Roles & Responsibilities (Continued)

GE Medical Systems is responsible for the following tasks:

- For Slab On Grade Sites: GE Installation Specialist to make sure the Magnet Floor Preparation Kit shall be installed, anchors pulled tested, and integrated to RF Shield prior to magnet delivery to allow time to address any issues that may arise.
- For Suspension Foundation Sites: GE Installation Specialist to make sure the Magnet Mounting Plate shall be installed, anchored, epoxy cured, integrated to RF Shield, and plate to floor gap filled-in prior to magnet delivery to allow time to address any issues that may arise.
- GE Installation Specialist to provide magnet mounting plate drawings and installation instructions to the RF Shield and mechanical contractor/rigger.
- GE Installation Specialist to assist Mechanical Contractor/RF Shield Room vendor by locating magnet isocenter during layout of Magnet Room for mounting anchors.
- GE Installation Specialist to inspect and verify the Magnet Floor Preparation Kit or the Magnet Mounting Plate installation is correct prior to magnet delivery. Carefully inspect the electrical connectivity of the anchor/stud to the RF Shield and Magnet Mounting Plate to make sure anchors are properly installed.
- For Slab On Grade: GE Service to work with riggers to attach magnet to floor anchors (Magnet Floor Preparation Kit) per Magnet installation documentation requirements.
- For Suspension Foundation Sites: GE Service to work with riggers to attach magnet to magnet mounting plate per Magnet installation documentation requirements.

Note

If a Suspension Foundation site RF Shield Room is installed at floor level the magnet mounting plate must be utilized as the RF Shield under the Magnet and the magnet mounting plate must be mounted to the concrete. In cases where the RF Shield is located well below the magnet, i.e. in a combined RF and Magnetic shield design, then the magnet mounting plate will not be utilized as part of the RF Shield. Refer to *Direction 2268297 GE 0.7T OpenSpeed Magnet Mounting Plate Installation* for additional details and install procedure.

7-7-3 Slab On Grade Sites Physical Requirements

Slab On Grade sites require the Magnet mounted to the concrete floor. The Magnet Support Ring is bolted through utilizing foam isolation pads and a bracket is utilized to mount the lower Coldhead to the concrete floor to minimize image artifacts generated by magnet vibration.

Note

For Slab On Grade Sites the 0.7T Magnet **cannot** be hard bolted to the floor and must be installed using the provided Magnet Mounting Kit to meet the vibration specifications.

5. **Important!!** The Magnet Room floor must meet the requirements specified in Table 7-5 for the Magnet and Coldhead mounting areas.
6. The Floor Preparation Kit provides a full-size Magnet Support Ring Mounting Template for locating floor holes, expansion anchors, anchor setting tools, and materials for RF sealing of the holes. See Illustration 7-12 for an overview of the template

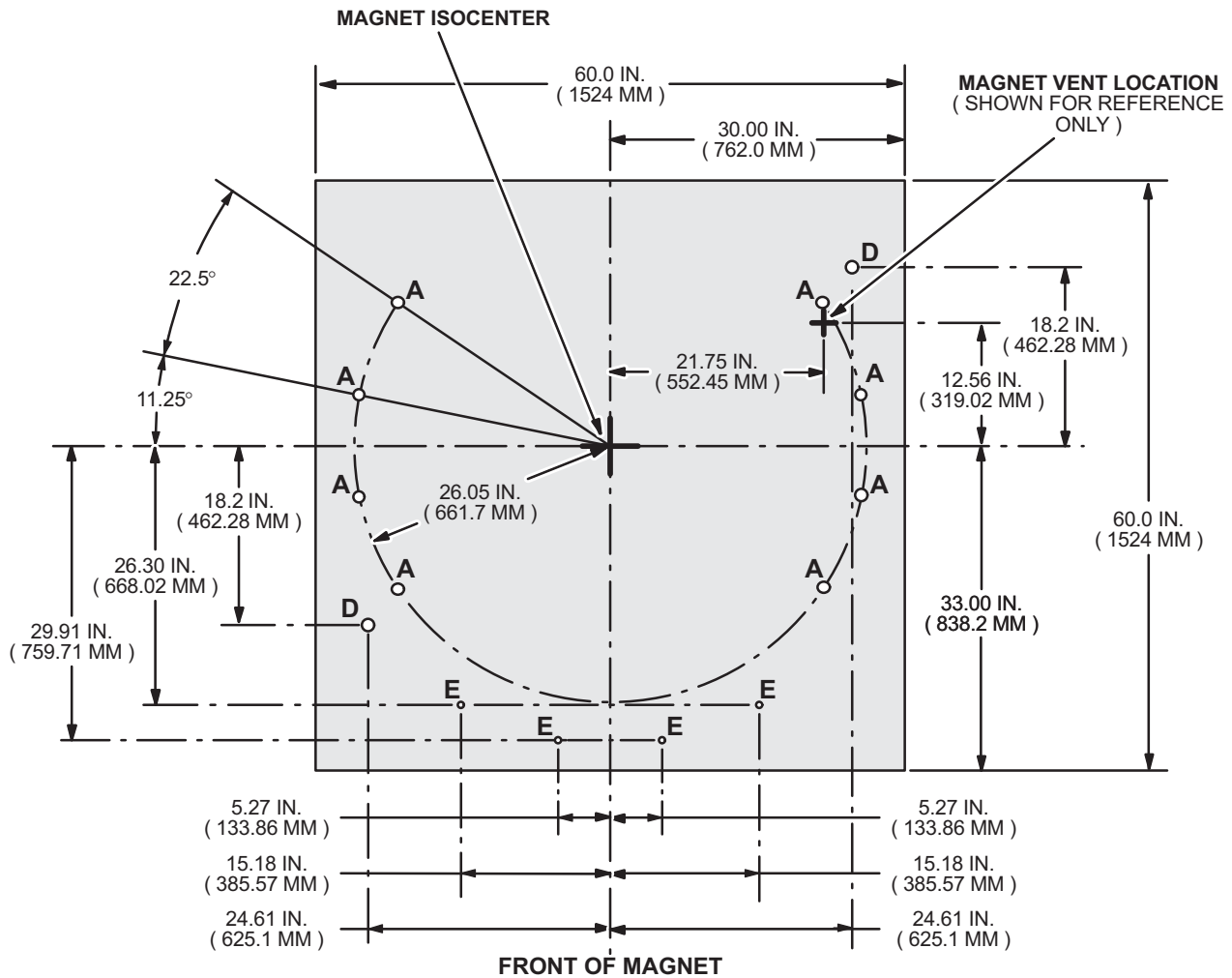
TABLE 7-5
FLOOR REQUIREMENTS FOR MAGNET AND COLDHEAD MOUNTING AREA

Floor Requirements	Specification
Flatness / levelness over mounting area	0.08 IN. (2 mm)
Minimum concrete depth	18 IN. (457 mm) integral with room and must be isolated from the rest of the building. Note Minimum concrete slab thickness must be within the entire floor area of the RF Shield Room and cannot be limited to the magnet footprint area only.
Floor surface	Continuous copper plate RF shield over concrete sub-floor. No finished floor material. No composite RF shield material. See Illustrations 7-13 and 7-14
Rebar / Metal Reinforcement	Rebar MUST be positioned per GE Foundation Drawing to avoid coldhead and crows.
ISO Center Marking	ISO Center crosshatch clearly marked on floor.

7-7-3 Slab On Grade Sites Physical Requirements (Continued)

NOTE:

- MOUNTING PLATE HOLE IDENTIFIERS:
A = USED TO BOLT MAGNET SUPPORT RING TO FLOOR.
D = USED FOR INSERTION OF ALIGNMENT PINS FOR MAGNET LOCATION.
E = TABLE RAIL MOUNTING HOLES



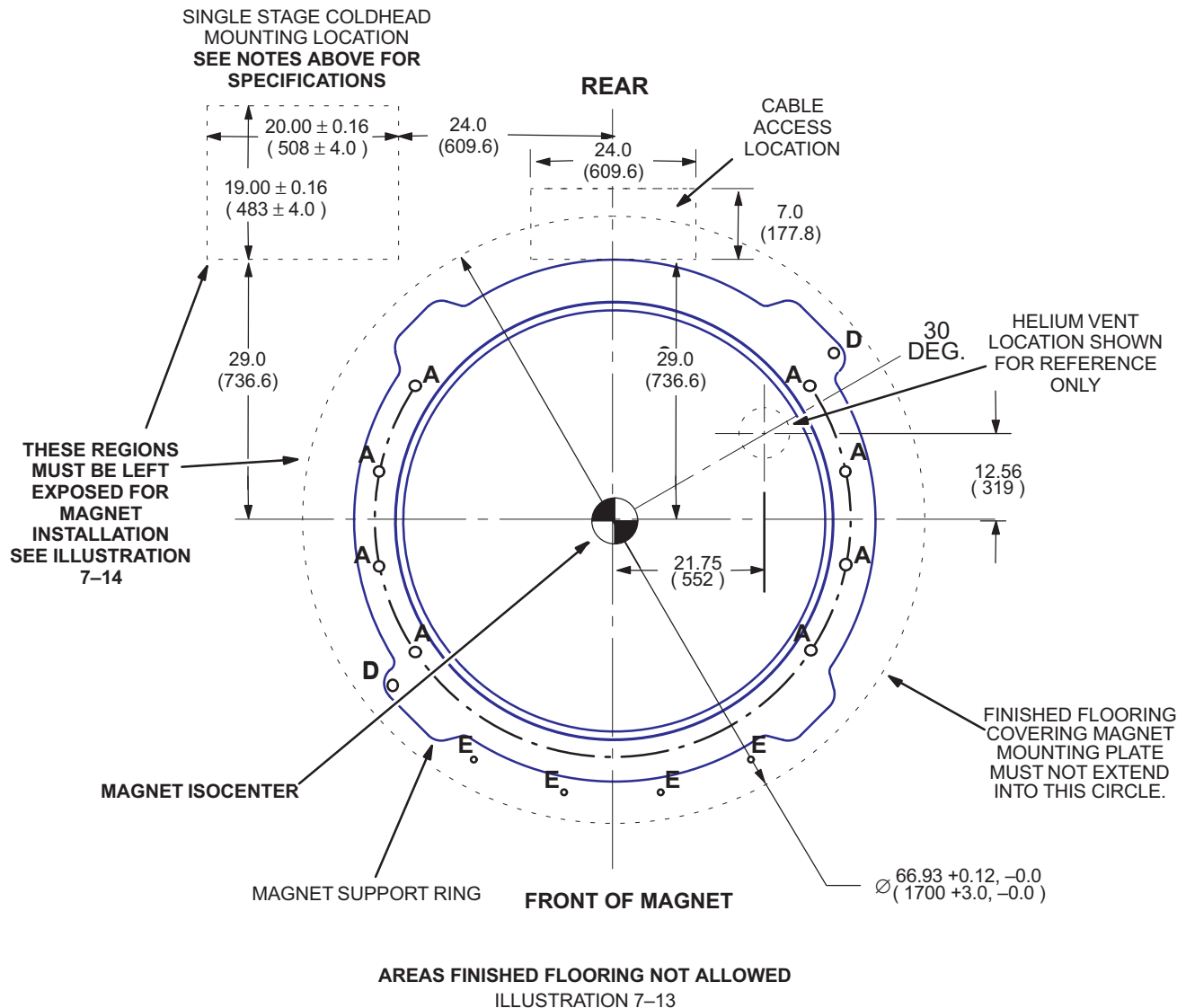
MAGNET SUPPORT RING MOUNTING TEMPLATE (2294577)

ILLUSTRATION 7-12

7-7-3 Slab On Grade Sites Physical Requirements (Continued)

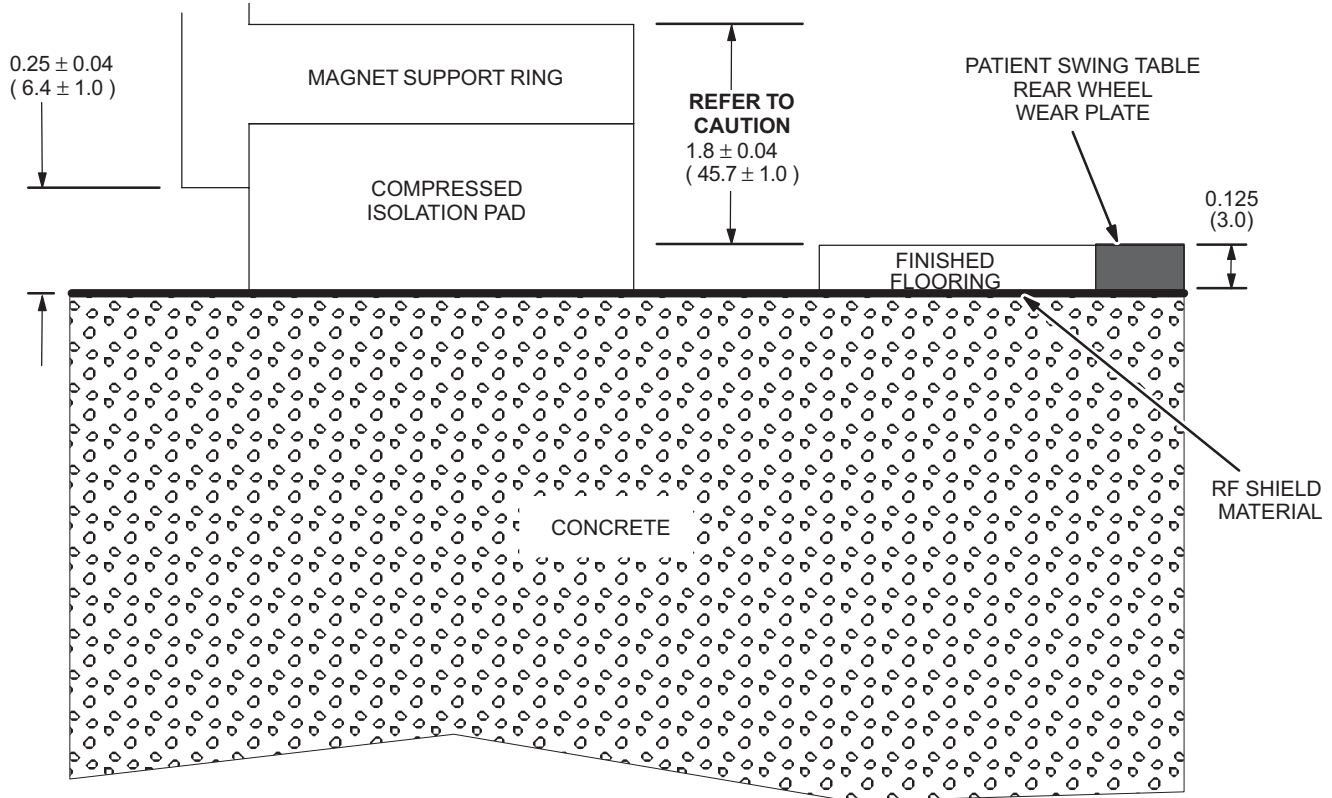
NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- **SINGLE STAGE COLDHEAD MOUNTING LOCATION FLOOR AREA MUST MEET FOLLOWING SPECIFICATIONS:**
 - SURFACE AREA TO BE AT SAME HEIGHT (WITHIN ± 0.5 INCHES (12.7 MM)) AND LEVEL (WITHIN 0.08 INCHES (2 MM)) AS MAGNET MOUNTING SURFACE OVER THE 20 INCH (508 MM) WIDE BY 19 INCH (483 MM) HIGH REGION SHOWN BELOW.
 - AREA TO HAVE MINIMUM 18 INCHES (457 MM) DEPTH OF CONCRETE WHICH IS INTEGRAL WITH THE MAGNET ROOM CONCRETE. THE COLDHEAD MOUNTING AREA SURFACE MUST NOT BE COVERED BY FINISHED FLOOR.
 - **KEEP FLUSH FLOOR TRENCH DUCT 3.0 INCHES (76 MM) OR MORE FROM COLDHEAD MOUNTING AREA.**
 - CONTINUOUS COPPER PLATE RF SHIELD OVER CONCRETE SUB-FLOOR. **NO** FINISHED FLOOR MATERIAL. **NO** COMPOSITE RF SHIELD MATERIAL.
 - REBAR MUST BE POSITIONED PER GE FOUNDATION DRAWING TO AVOID COLDHEAD ANCHORS.



7-7-3 Slab On Grade Sites Physical Requirements (Continued)

NOTE: ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



CAUTION

Shown dimension from top of Magnet Support Ring to top of finished flooring material & Patient Table Rear Wheel Wear Plate **MUST** be maintained for proper operation/alignment of the Magnet & Patient Table.

MAGNET SUPPORT RING & FINISHED FLOORING HEIGHT REQUIREMENT
ILLUSTRATION 7-14

7-7-4 Seismic Zone & Suspension Foundation Sites Physical Requirements

Important!! If a Suspension Foundation site RF Shield Room is installed at floor level the Magnet Mounting Plate must be utilized as the RF Shield under the Magnet. In cases where the RF Shield is located well below the magnet, i.e. in a combined RF and Magnetic shield design, then the magnet mounting plate will not be utilized as part of the RF Shield. Following are the requirements for utilizing the Magnet Mounting Plate.

1. The Magnet Mounting Plate location and orientation must meet centerline and vent location requirements for the room. Centerline and vent location are marked on plate. The plate location (recessed area) must be flat and level within 0.08 inches (2.0 mm) over the entire surface. The epoxy grout must be uniformly spread 0.20 – 0.25 inch (5.08 – 6.35 mm) thick in the grout area shown in Illustration 7-16 to result in the plate being level within 0.08 inch (2.0 mm) at all four corners. See Illustration 7-15.



RF Shield material thickness and finished flooring material thickness must be included in the determination of depth of Magnet Mounting Plate recess into the floor.

2. The Magnet Mounting Plate must be recessed into the floor in the area shown inside the dotted lines in Illustration 7-15. The remainder of room floor built up above the top of the plate in conformance with Table 7-6. A minimum depth of 18 inches (457 mm) of concrete is required in the Magnet Room. The minimum concrete depth under the mounting plate is 15.0 inches (381.0 mm) due to plate depression into the concrete.

Note

For Suspension Foundation Sites the 0.7T Magnet **must** be hard bolted to the Magnet Mounting Plate and must be installed using the provided Magnet Mounting Kit to meet the vibration specifications.

TABLE 7-6
FLOOR RECESS / HEIGHT REQUIREMENTS FOR MAGNET MOUNTING PLATE

Concrete Floor Plate Recess Depth From Top of Concrete to Top of Finished Floor (assumes 0.12 inch [3.0 mm] compressed grout thickness)	Finished Floor Height Above Top of Mounting Plate
2.25 inches $\pm$ 0.08 inches (57.1 mm $\pm$ 2.0 mm)	0.13 inch $\pm$ 0.04 inch (3.2 mm $\pm$ 1.0 mm)

3. Where the RF Shield Room is installed at floor level the magnet mounting plate must be an integral part of the magnet room RF integrity. The RF shield material must extend under the full perimeter of the mounting plate for a distance of 1.0 inch (25.4 mm) minimum to 1.5 inch (38.1 mm) maximum. RF shield material must be insulated from the concrete floor.
4. Floor rebar must be positioned within as shown on GE Foundation Drawing to prevent interference with mounting bolts for the "B" and "C" holes. See Illustration 7-15.

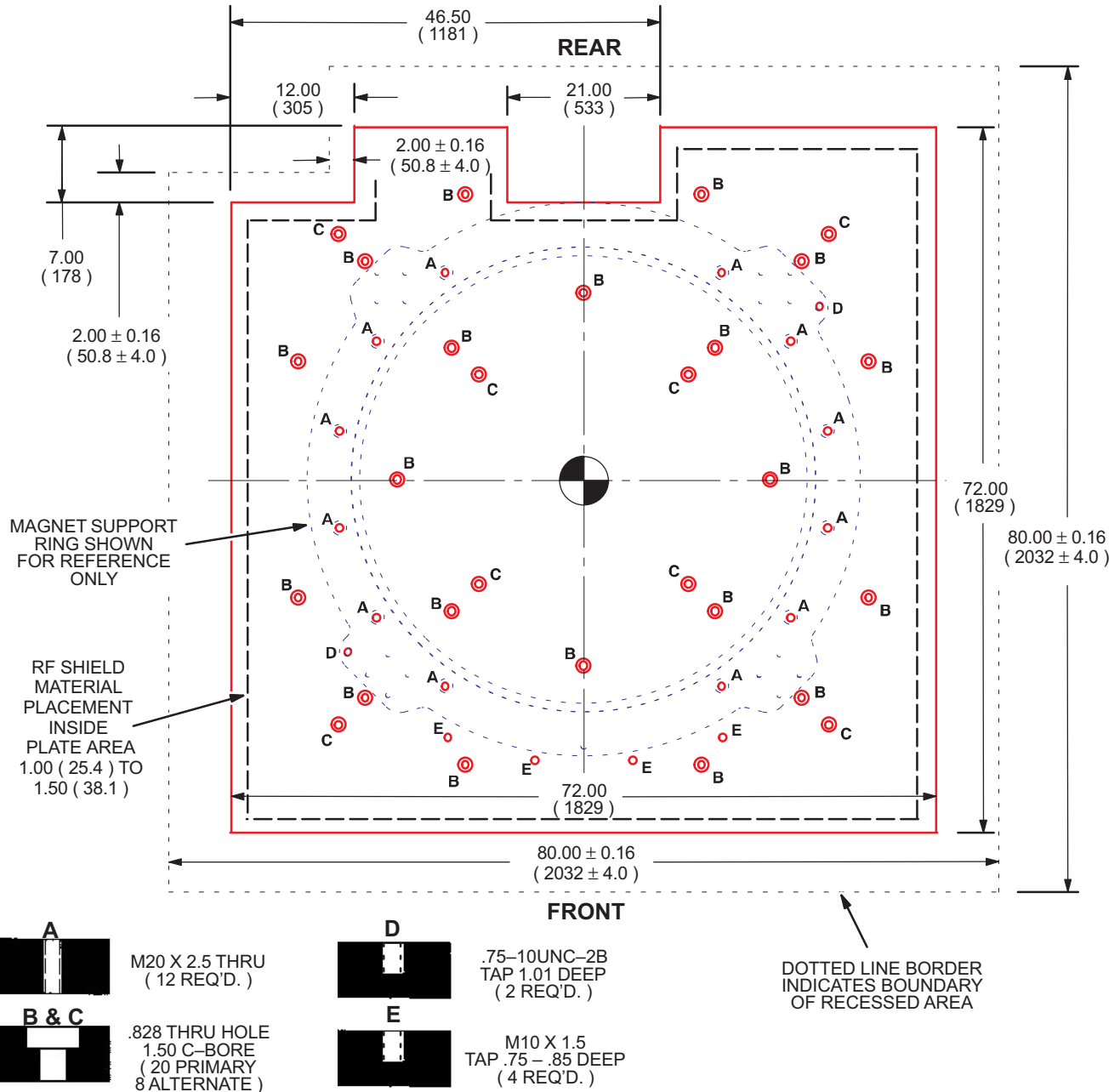
7-7-4 Seismic Zone & Suspension Foundation Sites Physical Requirments (Continued)

5. The top surface of the magnet mounting plate must be fully exposed under the circle of the magnet to allow for placement of the Magnet Mounting Kit pads between the magnet support ring and the magnet mounting plate. The concrete floor surface must also be left exposed (i.e. no material over or under the RF shield material other than insulation that must be under the RF shield material) in the Single-Stage coldhead mounting and cable trough locations. See Illustrations 7-15 through 7-19.
 6. The floor area where the Single-Stage Coldhead will be mounted must meet the following specifications:
 - a. Entire floor area must be composed of concrete integral with concrete to which Magnet Mounting Plate is attached. This concrete must **not** be finished with any subflooring / flooring materials prior to mounting Coldhead. Coldhead cannot be mounted through materials other than RF Shield and Conductive Pad specified in the Direction 2243897 GE 0.7T OpenSpeed Magnet Delivery and Installation.
 - b. Surface area to be the same height as Magnet Mounting Plate within ± 0.5 inches (12.7 mm) and level within 0.08 inches (2 mm).
 - c. Area to have a minimum 18 inches (457 mm) depth of concrete which is integral with the Magnet Room concrete. The coldhead mounting area surface must not be covered by finished floor.
- d. Rebar must be positioned per GE Foundation Drawing to avoid Coldhead anchors. See Illustration 7-17.

7-7-4 Seismic Zone & Suspension Foundation Sites Physical Requirements (Continued)

NOTE:

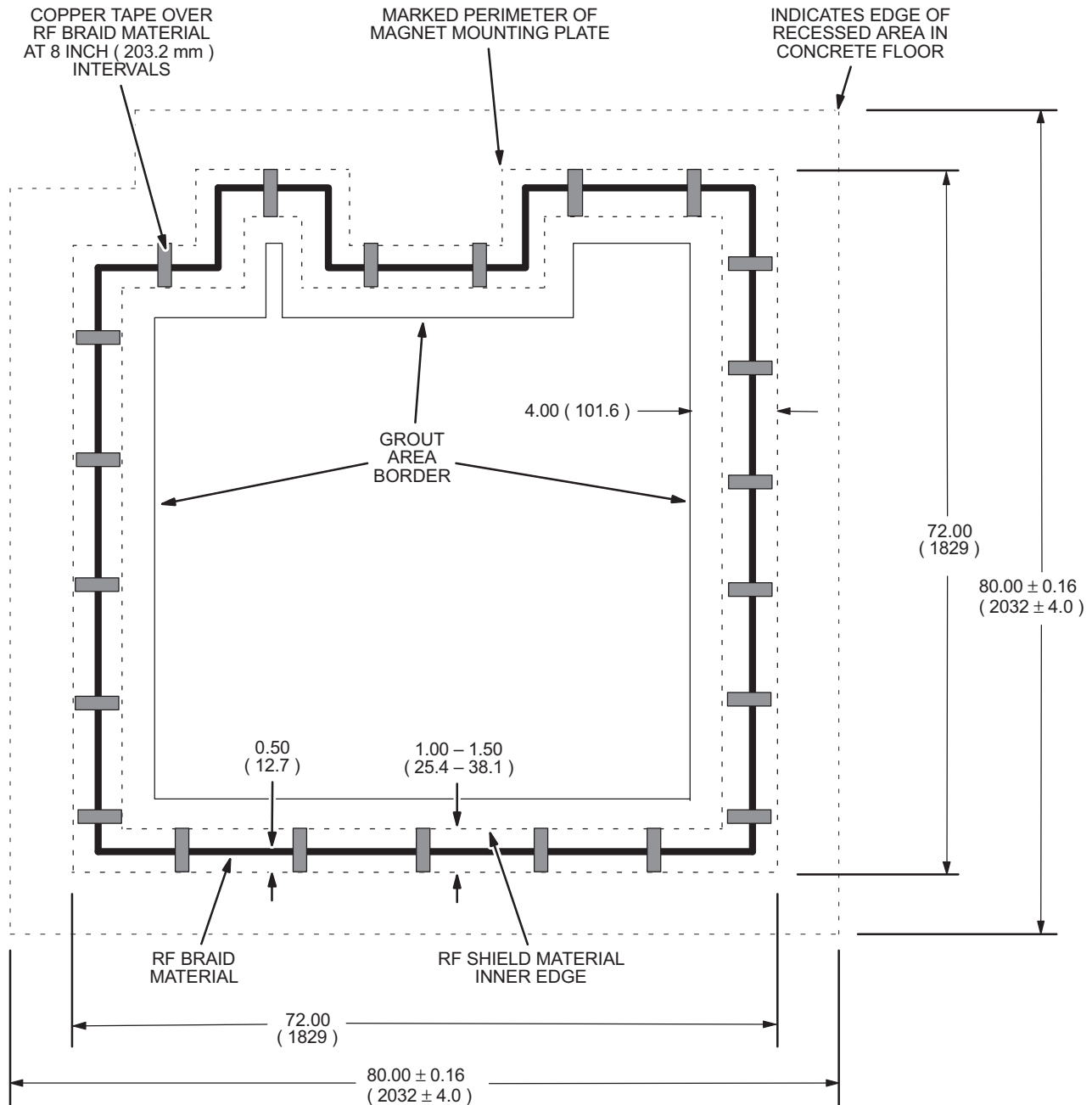
- * ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- * APPROXIMATE WEIGHT OF MAGNET MOUNTING PLATE IS 3,100 LBS (1409 KG).
- * MOUNTING HARDWARE AND MATERIALS ARE PROVIDED WITH GE SUPPLIED MAGNET MOUNTING PLATE.
- * MOUNTING PLATE HOLE IDENTIFIERS: **A** = USED TO BOLT MAGNET FLANGE TO MOUNTING PLATE. **B** = PRIMARY HOLES TO ATTACH MOUNTING PLATE TO BUILDING FLOOR. **C** = ALTERNATE HOLES TO ATTACH MOUNTING PLATE TO BUILDING FLOOR. **D** = USED FOR INSERTION OF ALIGNMENT PINS FOR MAGNET LOCATION. **E** = TABLE RAIL MOUNTING HOLES



0.7T MAGNET MOUNTING PLATE
ILLUSTRATION 7-15

7-7-4 Seismic Zone & Suspension Foundation Sites Physical Requirments (Continued)

NOTE: ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.

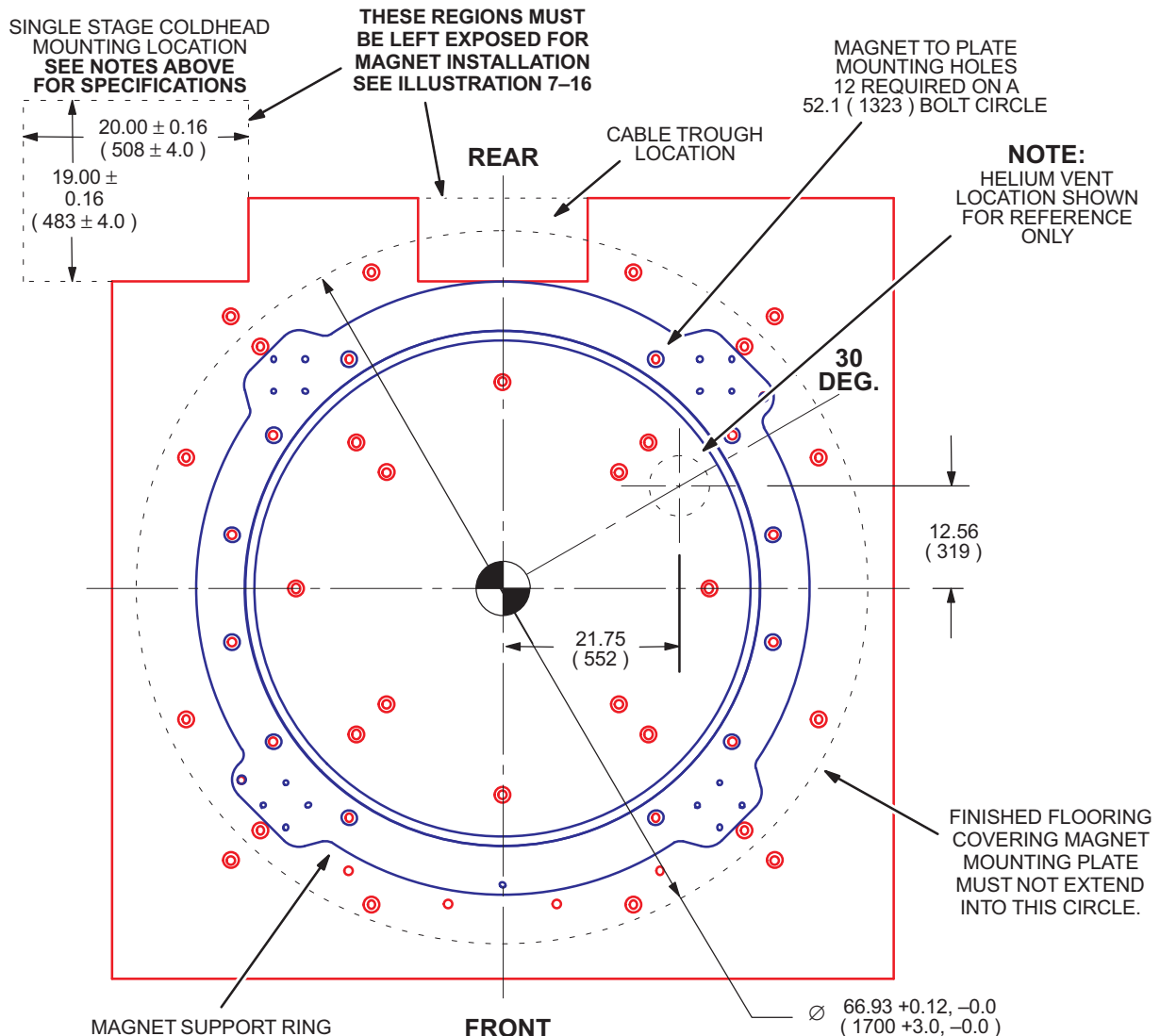


CONCRETE FLOOR PREPARATION FOR MOUNTING PLATE
ILLUSTRATION 7-16

7-7-4 Seismic Zone & Suspension Foundation Sites Physical Requirements (Continued)

NOTE:

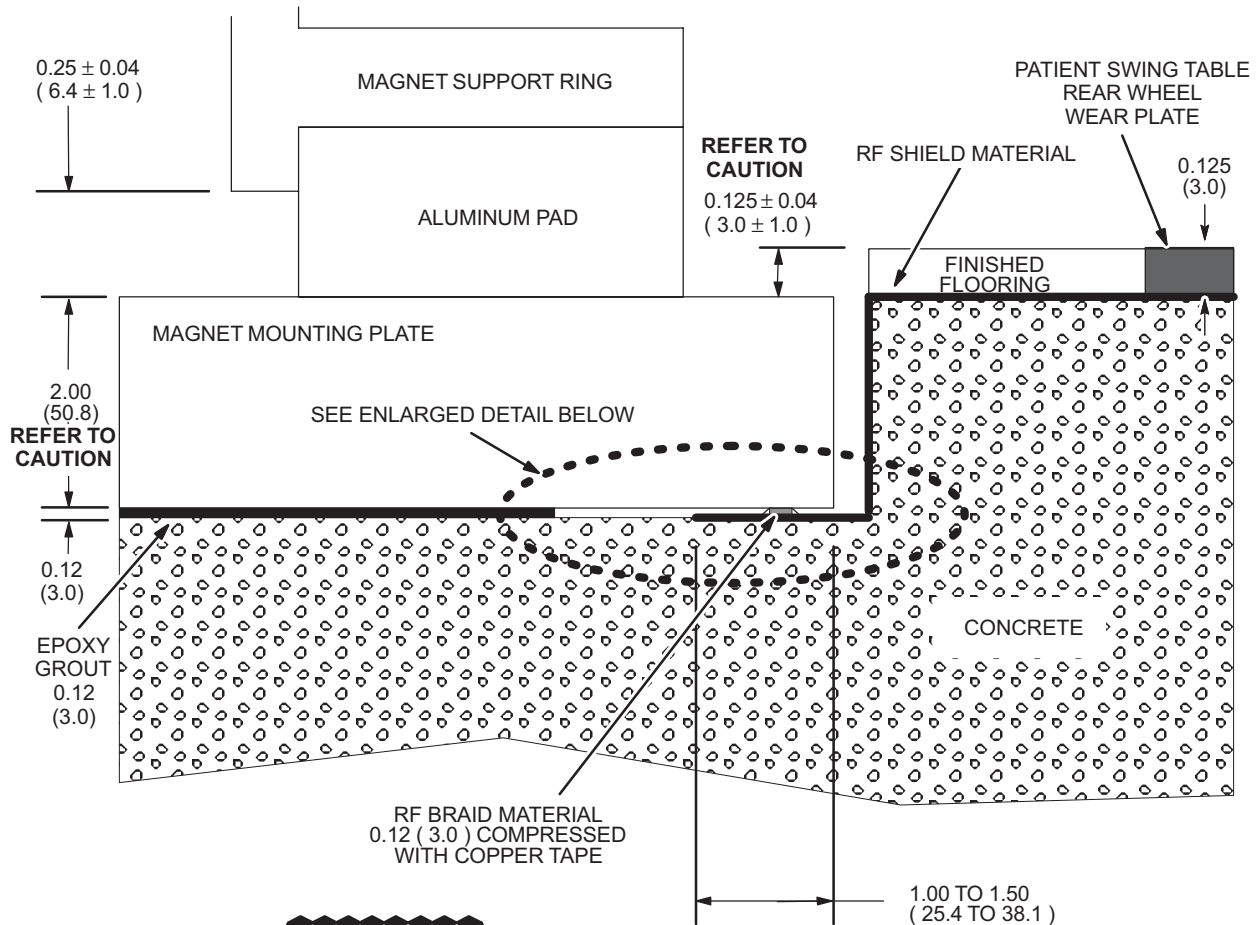
- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- **SINGLE STAGE COLDHEAD MOUNTING LOCATION FLOOR AREA MUST MEET FOLLOWING SPECIFICATIONS:**
 - SURFACE AREA TO BE AT SAME HEIGHT (WITHIN ± 0.5 INCHES (12.7 MM)) AND LEVEL (WITHIN 0.08 INCHES (2 MM)) AS MAGNET MOUNTING PLATE OVER THE 20 INCH (508 MM) WIDE BY 19 INCH (483 MM) HIGH REGION SHOWN BELOW.
 - AREA TO HAVE MINIMUM 18 INCHES (457 MM) DEPTH OF CONCRETE WHICH IS INTEGRAL WITH THE MAGNET ROOM CONCRETE. THE COLDHEAD MOUNTING AREA SURFACE MUST NOT BE COVERED BY FINISHED FLOOR.
 - **KEEP FLUSH FLOOR TRENCH DUCT 3.0 INCHES (76 MM) OR MORE FROM COLDHEAD MOUNTING AREA.**
 - REBAR MUST BE POSITIONED PER GE FOUNDATION DRAWING TO AVOID COLDHEAD ANCHORS



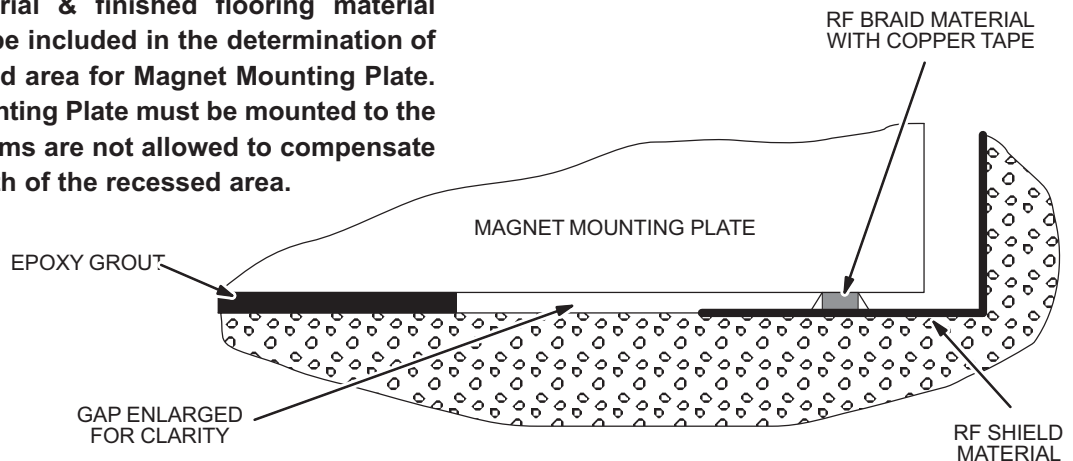
0.7T MAGNET FINISHED FLOOR REQUIREMENTS FOR MOUNTING PLATE
ILLUSTRATION 7-17

7-7-4 Seismic Zone & Suspension Foundation Sites Physical Requirements (Continued)

NOTE: ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



RF Shield material & finished flooring material thickness must be included in the determination of depth of recessed area for Magnet Mounting Plate. The Magnet Mounting Plate must be mounted to the concrete and shims are not allowed to compensate for errors in depth of the recessed area.



MAGNET MOUNTING PLATE INSTALLATION SIDE VIEW

ILLUSTRATION 7-18

7-7-4 Seismic Zone & Suspension Foundation Sites Physical Requirments (Continued)

NOTE: ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.

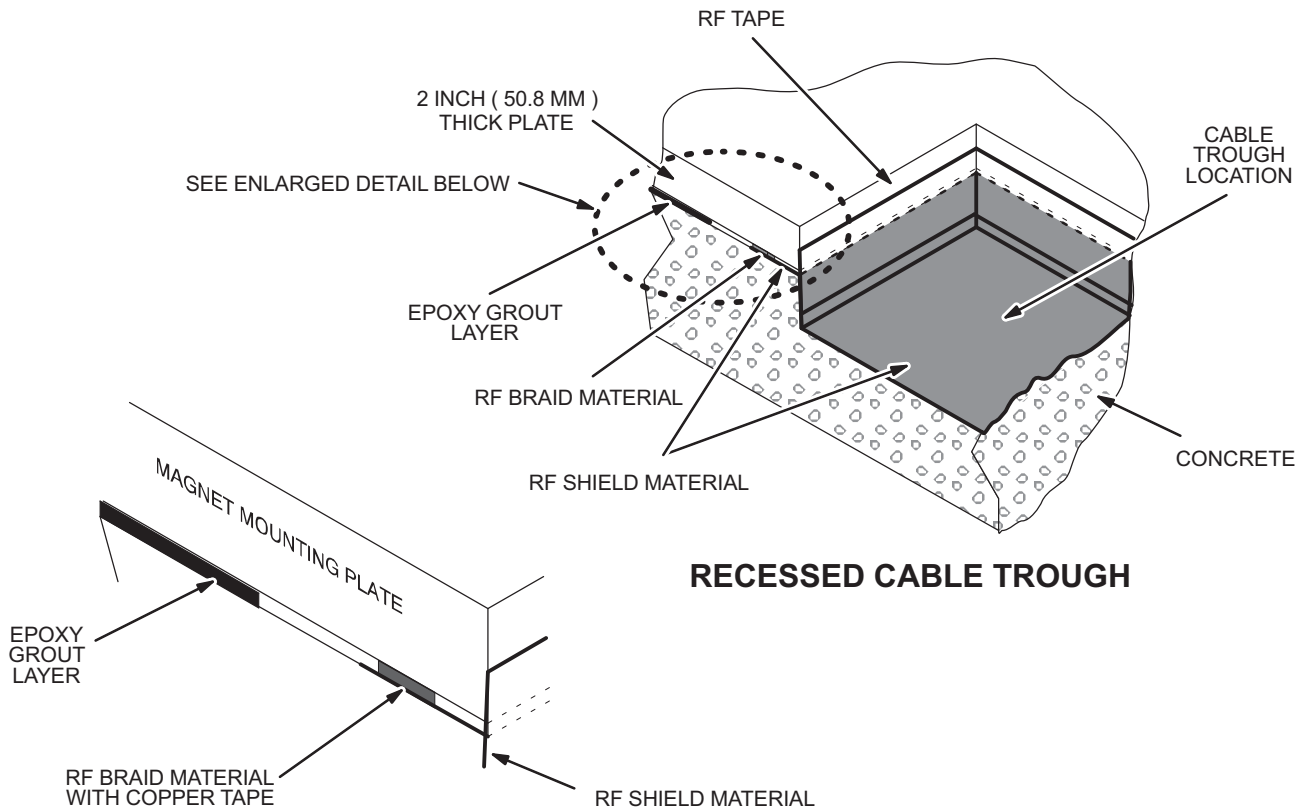
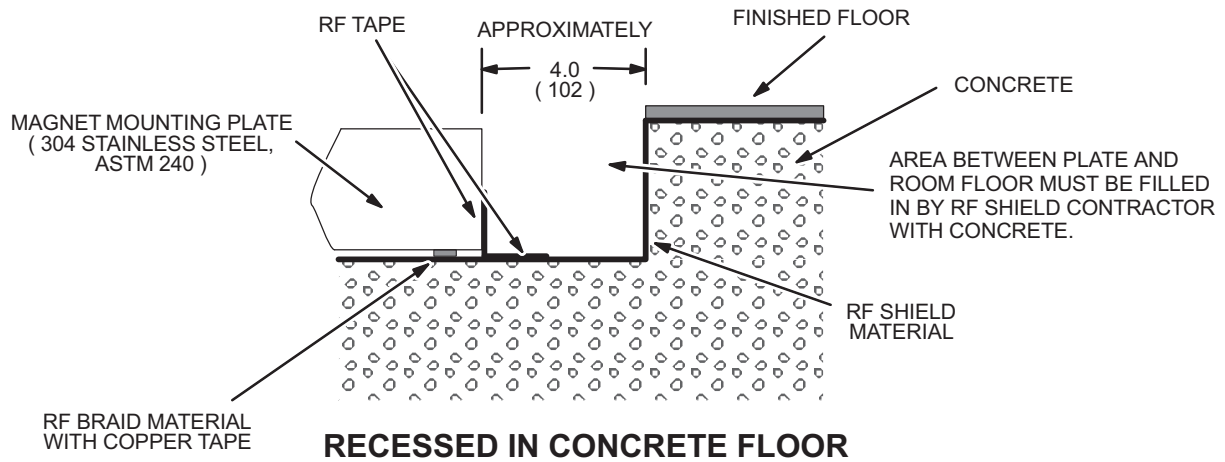


ILLUSTRATION 7-19

7-7-5 RF Shield Integrity

The anchor hardware must maintain RF shield integrity. This is accomplished by electrically sealing the stud at the penetration point on the RF shield. The method by which the electrical contact is made must take into account any stretch in the stud resulting from the applied clamping force (tension). RF tape applied over magnet mounting plate holes will minimize the ingress of debris (i.e. metal filings) in the holes which could introduce artifacts into the MR images. The RF room test should result in a specific attenuation at the operating frequency of the system under the following conditions:

1. Blank Penetration Panel installed
2. Floor Preparation Kit or Magnet mounting plate installed and anchored.
3. Electrical connection made between the anchor stud and the RF shield.

Refer to APPENDIX A – RF SHIELDED ENCLOSURE TEST GUIDELINE.

7-7-6 Electrical Isolation

The Magnet mounting hardware must not provide a secondary ground path for the RF Shield Room. A secondary ground may occur by having the anchor come in contact with steel rebar, wire mesh or structural steel in the floor. Ideally the ground impedance between the anchor and earth ground should be greater than 1000 ohms. This may not be possible due to moisture conditions in the concrete or soil beneath the concrete. Therefore the following requirements for ground isolation shall be observed.

1. A ground isolation test performed on each electrically conductive anchor and stud.
 - a. If the result is greater than 1000 ohms on each stud then record the resulting measurement and give documentation to GE Medical Systems.
 - b. If the result is less than 1000 ohms but greater than 100 ohms, contact the power and grounding support personnel at GE Medical Systems and review process and site conditions.
 - c. If the result is less than 100 ohms then it is very likely the anchor has made contact to steel rebar or wire mesh. In this case the steel in the floor will need to be removed or the anchors will need to be relocated. In either case GE Medical Systems must be notified and a retest performed after the corrective action is taken.
2. The test results must be recorded by RF Shield Room Vendor and the information forwarded to GE Medical Systems for the customers record file.

7-7-6 Electrical Isolation (Continued)

Electrical Isolation Measurement Method

Prior to attaching the primary ground and installing any power outlets, room lights, water supplies, etc. into the RF Shield Room, a ground isolation test is performed between the room's ground plane and earth ground.

- This measurement should be greater than 1000 ohms DC. However, the results of this measurement may appear lower for RF Shield Rooms located below or at grade level. This is caused by a capacitive voltage, setup by the measurement meter's DC voltage, between the RF Shield Room and earth ground.
- For RF Screen Rooms with a resistance reading less than 1000 ohms the following method should be used to determine if the low resistance reading is caused by this capacitive effect.
 - Use a Meggar Insulation Tester capable of reading < 1000 ohms to most accurately make the measurement. An analog meter with a D'Arsonval meter movement may be used if a Meggar is not available.
 - After making the measurement reverse the leads and watch for the measurement to start high and decrease to a lower resistance value. This change in measurement verifies the capacitive effect.
 - In this case the peak measurement is the approximate resistance between the RF Shield Room and earth ground.

Note

For sites where magnet mounting is isolated from the RF Shield (RF Shield positioned below concrete) the magnet will require grounding to the RF Shield.

7-7-7 Magnet Mounting Plate Anchors Installation Location

For Slab On Grade Sites the exact locations for installing the Magnet Support Ring anchors are determined by utilizing the Floor Preparation Kit template, see Illustration 7-12.

For Seismic Zone and Suspension Foundation Sites the exact locations for installing the magnet mounting plate anchors are determined by Illustration 7-15. The anchors will be installed after the magnet mounting plate is in position on the floor. Refer to *Direction 2268297 GE 0.7T OpenSpeed Magnet Mounting Plate Installation* for additional details and install procedure.

Coordination between the RF Shield Room Vendor and Building Contractor/Architect may be necessary to mark the location of the anchors to prevent interference with rebar or structural steel. A re-arrangement of the room layout may be necessary to ensure ground isolation.

7-8 MAGNET ROOM – WALL MOUNTED EQUIPMENT REQUIREMENTS**7-8-1 Remote Oxygen Sensor Module (OM3) – Optional**

The Remote Oxygen Sensor Module (if option ordered) must be mounted approximately 60 in. (1524 mm) above the Magnet Room floor near the front of the magnet enclosure.

7-8-2 Magnet Rundown Unit (MS4)

The magnet rundown unit should be mounted 60 in. (1524 mm) above the Magnet Room floor near the front of the magnet enclosure but outside the 200 gauss zone.

7-8-3 Emergency Off buttons

Emergency off buttons supplied with the Main Disconnect Panel (MDP) are to be located near room exit of magnet and equipment rooms. These buttons must be clearly labeled, "Emergency Off". Refer to Section 5, POWER REQUIREMENTS, for emergency off power requirements.

7-9 RF DOOR SWITCH

RF shielded room vendor must supply and install RF door switches on all RF shielded doors. These switches must be wired in series and a GE supplied cable (two loose lead conductors) will attach to one door switch. RF switches must be rated for 24 volts at 750 milliamperes maximum and the switches must be in the open position when the doors are open (switch contacts close when the doors are completely closed).

7-10 EMERGENCY EXIT

Emergency exiting from the Magnet Room is to be specified by the customer's architect and contractor. Such measures as an out swinging door, emergency door latch release, easily removed window, or other measures must be designed into the room. Emergency exit instructions must be permanently and prominently mounted near the door and/or window.

7-11 ROOM VENTILATION SWITCH

Placement of the room ventilation switch should be near the Magnet Room door and is the responsibility of the architect and mechanical contractor.